BFGS on Convex Nonsmooth Functions: Convergence and Failure

Author-review edition, 15 October 2026

Abstract Ordinary full BFGS often succeeds on nonsmooth objectives, but its convergence theory must accommodate gradient jumps and possible degeneration of the quasi-Newton matrices. We give complementary positive and negative results for finite convex objectives with Armijo–weak-Wolfe steps and actual gradients at differentiable accepted iterates. A centered smooth–sublinear theorem proves global whole-sequence convergence for $u^2 + |v|$ and applies in arbitrary dimension on compact initial sublevels contained in the region of the centered representation. Coercive degree-one homogeneous objectives, including coercive max-linear functions, norms and gauges, admit global convergence. Local results cover finite polyhedral smooth-piece partitions and nondegenerate planar smooth maxima; in an explicit curved-interface example, moving comparison geometry proves convergence although the radial criterion fails. In contrast, a fixed unit-first implementation can converge to nonstationary points from starts arbitrarily close to the unique minimizer of one fixed objective. One construction is a maximum of four globally convex $C^{2,1}$ branches; another retains uniform strong convexity with $C^{1,1}$ branches, three of which lose second differentiability at the minimizer. The positive proofs derive an adjacent-degree hierarchy for actual principal minors of full-BFGS matrices, then combine reciprocal growth with comparison energies. The resulting objective-level conditions require neither iterate convergence nor prescribed spectral behavior. The negative proofs certify the complete trial trees and infinite local embeddings. These results expand the convergence theory while ruling out unrestricted deterministic guarantees for the specified classes.

Keywords BFGS · convex nonsmooth optimization · global convergence · local convergence · counterexample

Mathematics Subject Classification (2020) 90C30 · 90C53 · 49J52

1 Introduction

BFGS builds its search directions from changes in gradients, yet it often works well when those gradients jump. Classical convex convergence theory uses smoothness together with the evolving quasi-Newton matrices [7,8,9]. Lewis–Overton's experiments and analyses [1] exposed the difficulty of extending that theory: nonsmooth BFGS can approach a minimizer while its inverse-Hessian

approximations become highly singular. What global or local convergence can ordinary full BFGS guarantee on convex nonsmooth functions, and how can one prove it without assuming the asymptotic behavior of its iterates or matrices?

We give three answers: deterministic counterexamples to every-start local convergence; global and local convergence theorems for structured convex families, including the mixed benchmark below; and a principal-minor and comparison-energy argument that remains effective as the BFGS matrices degenerate.

Lewis–Overton's Challenge 7.1 [1] gives this question a broader, probabilistic setting. For locally Lipschitz semialgebraic objectives with bounded sublevels and absolutely continuously distributed initial data, it asks about nonbreakdown, stationary cluster points, R-linear convergence of all function trial values, and identification of spectral subspaces under partly smooth convergence. We study deterministic guarantees for convex objectives. Successful termination is allowed; infinite runs are analyzed under the differentiable-iterate convention made precise below.

Substantial positive theory precedes this work. Guo–Lewis extends Powell's argument through smooth convex replacements on regions relevant to a BFGS sequence [5, Theorems 3.2, 4.1 and 4.3]. The resulting theory proves Euclidean-norm convergence in arbitrary finite dimension [5, Corollary 4.2]; for the mixed model

$$F(u, v) = u^2 + |v|,$$

Corollary 4.4 gives only a subsequence approaching a point of $v = 0$. The paragraph after that corollary explicitly leaves convergence of every BFGS sequence to $(0, 0)$ open [5, Corollary 4.4 and following discussion]. Approaching the nonsmooth line does not force the smooth coordinate u to vanish. Corollary 2 answers this whole-sequence question for every admissible full-BFGS run, as a consequence of an arbitrary-dimensional family theorem rather than a special planar recurrence.

Two other advances clarify the remaining task. Xie–Wächter's two-piece unbounded affine model leads to an unbounded descent direction after finitely many iterations (2017 preprint, arXiv:1712.08571v1, Theorem IV.1); the value-convergence theorem for a broader homogeneous family in Chapter 7 of the complete companion (introduced at the end of this section) does not replace that finite escape result. Gebken's published 2026 introduction still identifies $u^2 + |v|$ as unresolved. His local criticality theorem for continuous smooth selections assumes an iterate limit, repeated activity, and specified spectral and gradient-jump behavior [3, Theorem 3.1 and (B1)]. Our positive results instead derive convergence from objective hypotheses within the convex families stated here; they do not subsume his potentially nonconvex selection setting.

Throughout the paper, an *admissible run* uses the unmodified full BFGS update, an arbitrary $H_0 \succ 0$, fixed $0 < c_1 < c_2 < 1$, Armijo–weak-Wolfe steps, and differentiable accepted points with nonzero gradients (Definition 1). The positive theorems concern every infinite such run. They do not assert that every search ray contains an acceptable differentiable point.

Deterministic limits. Prescribed-orbit constructions give smooth nonconvex failures [14,15]; nonconvexity also separates the examples of Lewis–Zhang [12] and Diao (2026 preprint, arXiv:2609.09686v1) from the question here. Lewis–Overton's nonsmooth examples use exact search or subgradient choices (*Behavior of BFGS with an Exact Line Search on Nonsmooth Examples*, 2008 manuscript), while Asl–Overton's failure theorem concerns a scaled memoryless update [13]. These cited examples do not supply convex local failure with the full update and the specified search. We construct convex objectives for which a fixed unit-first Armijo–weak-Wolfe doubling/bisection search generates full-BFGS iterates converging to a nonstationary point. For each local construction, one fixed objective has failing starts arbitrarily close to its unique minimizer, all with $H_0 = I$, and every accepted or rejected query approaches that minimizer across the family of runs. One construction is a maximum of four convex $C^{2,1}$ branches with bounded Hessians; another is a maximum of four uniformly strongly convex $C^{1,1}$ branches, three of which are not twice differentiable at the minimizer (Theorems 10 and 11). These refute every-start neighborhood guarantees under either regularity regime. They establish no positive-measure failure set and do not refute the almost-sure semialgebraic challenge.

Global and local convergence. We prove global value convergence for centered smooth-plus-sublinear objectives with compact initial sublevels contained in the region where the representation holds, and whole-iterate convergence when the minimizer is unique (Theorem 5). This settles the mixed benchmark above. For coercive finite convex degree-one homogeneous functions, including coercive maxima of finitely many linear functions, every admissible run converges to zero (Theorem 6). This includes every norm, for example $\|x\|_1$, whose nonsmooth set for $n \geq 2$ contains coordinate hyperplanes, beyond the isolated nonsmooth point of the Euclidean-norm case above. Local results cover smooth convex plus finite polyhedral terms and finite polyhedral partitions with smooth extensions (Theorem 7). Moving comparison geometry also yields local convergence for planar smooth maxima under strict-contour and active-gradient nondegeneracy assumptions (Corollary 7). The explicit three-quadratic example in Theorem 9 has curved switching sets and fails the radial condition, so this extension has a concrete payoff beyond inward radial modeling.

Analysis with degenerating matrices. The classical smooth analysis combines trace and determinant identities in a matrix potential [7, Section 5]. The uniform smooth secant bound used there need not hold when gradients jump. We prove an inequality linking each principal-minor degree to the next for the actual full-BFGS update. This adjacent-degree control forces the arithmetic mean of the reciprocals $1/(g_k^T H_k g_k)$ to diverge when gradients are bounded and predicted decreases tend to zero. Comparison energies turn this restriction into convergence: a field must retain information about the objective gap, control the gradient changes it sees, and move with a controlled energy cost. The field is an analytical device, not part of the algorithm. Neither a bounded condition number nor a prescribed spectral limit is needed (Theorems 1–3).

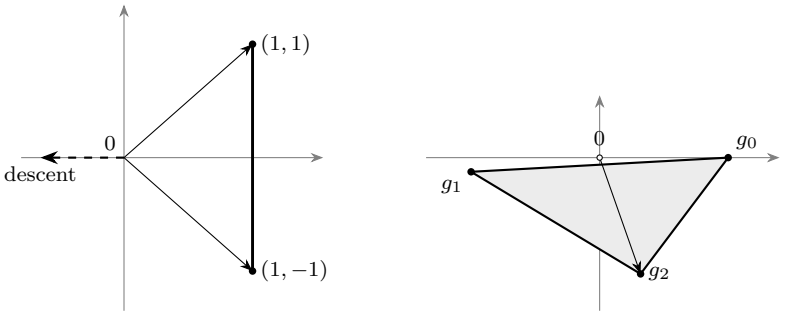
Our conditions are sufficient; they do not classify every successful run. Sections 2 and 3 explain and establish the mechanism; Sections 4 and 5 prove its global and local consequences, Section 6 gives the deterministic counterexamples, and Section 7 lists the remaining questions. The article includes the essential proofs of these claims. An independent complete companion preserves the wider noncompact and quantitative theory, with Lean proofs, exact correspondence and reproducible audit evidence.

2 What the results say about nonsmooth BFGS

The mixed ridge $F(u, v) = u^2 + |v|$ isolates the difficulty left by a subsequence approaching the nonsmooth line. Reaching $v = 0$ does not eliminate the horizontal error u^2. Yet the vertical gradient jump does not remove the direction that can reduce this error. Along $v = 0$ the limiting gradients are $(2u, 1)$ and $(2u, -1)$. Their difference is vertical. At a point with $u \neq 0$, the horizontal vector $(-u, 0)$ has the same strictly negative pairing $-2u^2$ with both gradients (Figure 1, left). The inward radial model also has a small error: for $v \neq 0$ and $0 \leq \tau \leq 1$,

$$F((1-\tau)(u, v)) - F(u, v) + \tau \nabla F(u, v)^T (u, v) = \tau^2 u^2. \tag{1}$$

Thus the nonsmooth part is exactly linear under inward scaling, while the smooth part contributes a quadratic remainder. Section 4 uses this property to prove convergence of the whole sequence. The inward segment is a comparison used in the proof; it is not a direction BFGS is assumed to take. No eventual eigenvector or eigenvalue description is needed.



(a) $u^2 + |v|$, $u = 1/2$ (b) Certified nonstationary seed

Fig. 1 Gradient space. Left: the mixed ridge at $u = 1/2$ has a vertical jump and a common horizontal descent direction. Right: the particular three-gradient seed used in Section 6. Its convex hull excludes zero, but its affine hull is the plane. The vertices on the right are rendered from the certified root's rational center; the mathematical definition uses the isolated root. Zero lies just above the edge $[g_1, g_0]$, at distance $1/\sqrt{257}$ from the hull. Solid arrows indicate gradients and the dashed arrow a common descent direction; none is a BFGS step.

Objective geometry	Starting scope	Conclusion
Centered smooth–sublinear, Theorem 5	Compact initial sublevel where the representation holds	Optimal values; unique minimizer gives whole-sequence convergence
Coercive degree-one homogeneous, including ℓ_1 and ℓ_∞ norms, Theorem 6	Every starting point	Whole sequence converges to zero
Finite polyhedral smooth-piece partition, including squared loss plus $\lambda\|x\|_1$ ($\lambda > 0$, unique minimizer), Theorem 7	Common low sublevel near the unique minimizer	Whole sequence converges to the minimizer
Nondegenerate planar smooth maximum, Corollary 7	Local basin under the stated contour hypotheses	Whole sequence converges to the unique minimizer

Table 1 What the positive theory guarantees. The benchmark $u^2+|v|$ has compact sublevels and a centered representation on all of $\mathbb{R}^2$, so the first row gives global convergence. The results concern admissible runs, without a presumed matrix limit.

The right panel of Figure 1 asks a different question: can a genuine full-BFGS run contract around a point where descent still exists? The bad corner is specific. Its slopes are $g_0 = (1, 0)$, $g_1 = (-1, -1/8)$ and $g_2 = (c_\circ, d_\circ)$, where $(c_\circ, d_\circ)$ is defined exactly in Appendix C. They are affinely independent, so no nonzero direction pairs equally with all three. Yet $(-1, 16)$ has negative pairing with each, and zero is outside their convex hull. The corner is therefore nonstationary. Section 6 constructs a three-step cycle and verifies that the stated line search produces it, including all rejected trials. This additional dynamical proof is what makes the corner a counterexample.

This distinction between affine and convex balance helps interpret the examples. For a nonempty finite family of slopes a_i, equality $a_i^T v = \kappa < 0$ for every i asks for a common rate of descent, whereas $a_i^T v < 0$ asks only for descent. The former fails when zero belongs to their affine hull; the latter fails when zero belongs to their convex hull. Indeed, a linear functional constant on the slopes is constant on their affine hull. Conversely, if zero is outside that affine hull, its nearest point gives a separating functional constant on the hull; changing its sign gives the negative common value. The convex version follows from the same closest-point argument for the convex hull. These are elementary separation facts, not a classification of BFGS behavior.

The positive conclusions differ in their geometry and in the locality of the guarantee. Table 1 distinguishes them before the proofs. In every row the quantifier is over infinite admissible runs with arbitrary positive-definite initialization. An admissible run uses actual gradients and satisfies Armijo–weak-Wolfe at its accepted points; the definition does not assert that an arbitrary line-search implementation always finds such points.

The first two rows make precise global assertions; they do not claim global convergence for every affine maximum. Indeed, the coercive polyhedral realiza-

tion of the wrong corner in Corollary 8 has a positive limiting objective gap. General affine offsets can place the bad corner away from the true optimizer, whereas homogeneity centers all linear pieces at the origin.

Local behavior raises a further distinction. Finitely many smooth formulas can reproduce bad configurations at infinitely many shrinking scales. A finite polyhedral partition near the optimizer controls the inward model because each remaining cell contains the optimizer after localization. Curved interfaces require a different argument: the planar example in Section 5 has a radial obstruction, yet a moving contour field proves convergence. These are complementary sufficient mechanisms and specific realized failures, rather than a classification by jump span.

Checking the hypotheses. Several analytical conditions are automatic once the objective geometry and the admissible-run convention hold. Weak Wolfe preserves positive-definite matrices and gives positive secant curvature. On a compact sublevel, finite convexity bounds the gradients; Armijo makes the predicted decreases summable. The substantive checks in Table 1 are therefore the centered representation, the finite local cells, or the planar switching geometry. For the centered route one must locate an actual minimizer and verify the decomposition on a region containing the initial sublevel. For the polyhedral route one checks the smooth cell extensions and uniqueness; the squared-loss plus ℓ_1 example in Section 5 makes that local decomposition explicit. For a planar maximum one checks the active gradient linear system at the minimizer: affine independence and strictly positive balancing weights. A common positive branch-Hessian lower bound supplies the strict-contour condition in Corollary 7.

Appendix E verifies these conditions jointly with actual full-BFGS histories. Proposition 2 specifies all three certificate inequalities for the mixed ridge and an infinite nonsmooth run; Proposition 3 gives a full-rank calibration in every dimension. Proposition 4 supplies a simple parameterized family with curved switching. The printed nonradial example also has infinite runs for every initial positive-definite matrix and valid constant pair (Corollary 9). The qualification on run existence is essential: Proposition 5 gives a coercive homogeneous objective and initial data for which no differentiable weak-Wolfe first step exists. These are sufficient classes with explicit witnesses, not a genericity or necessity claim for nonsmooth BFGS.

3 The analytical framework

Smooth convergence proofs can relate a gradient difference to curvature along a short step. A nonsmooth secant can instead contain a jump that remains large as the step shrinks. Our starting point is to measure how much of that secant is seen by a direction that still detects the objective gap. A quadratic comparison energy separates the cost of this component from a negative term measuring missed progress. The matrix hierarchy then shows that, when the

gradients are bounded, persistent missed progress cannot be paid for by a bounded secant cost per iteration. The argument uses the matrices actually generated by BFGS throughout.

Definition 1 (Admissible runs) Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, with $n \geq 1$. An admissible run is an infinite sequence of differentiable accepted points x_k, nonzero gradients $g_k = \nabla F(x_k)$, and matrices generated by

$$s_k = -\alpha_k H_k g_k, \qquad x_{k+1} = x_k + s_k, \qquad y_k = g_{k+1} - g_k,$$
$$H_{k+1} = \left(I - \frac{s_k y_k^T}{t_k}\right) H_k \left(I - \frac{y_k s_k^T}{t_k}\right) + \frac{s_k s_k^T}{t_k}, \qquad t_k = y_k^T s_k, \tag{2}$$

where $H_0 \succ 0$, $\alpha_k > 0$, and fixed constants $0 < c_1 < c_2 < 1$ satisfy

$$F(x_{k+1}) \leq F(x_k) + c_1 g_k^T s_k, \qquad g_{k+1}^T s_k \geq c_2 g_k^T s_k. \tag{3}$$

These are the usual actual-gradient, full-BFGS conventions [1]; see also (Xie and Wächter, 2017 preprint, arXiv:1712.08571v1). A zero gradient is successful termination. Statements about admissible runs do not assert that a differentiable acceptable point exists on every ray, or that every implementation finds one.

Write

$$B_k = H_k^{-1}, \qquad q_k = g_k^T H_k g_k, \qquad d_k = -g_k^T s_k = \alpha_k q_k,$$
$$\Delta_k = F(x_k) - F(x_{k+1}), \qquad \omega = 1 - c_2.$$

Lemma 1 *Every admissible run satisfies $H_k \succ 0$, $q_k, d_k, t_k > 0$ and*

$$c_1 d_k \leq \Delta_k \leq d_k, \qquad t_k \geq \omega d_k. \tag{4}$$

Moreover,

$$B_{k+1} = B_k - \frac{g_k g_k^T}{q_k} + \frac{y_k y_k^T}{t_k}, \qquad \widehat{B}_k := B_k - \frac{g_k g_k^T}{q_k} \succeq 0, \qquad \widehat{B}_k s_k = 0. \tag{5}$$

If the objective values have a finite limit, then $\sum_k d_k < \infty$.

Proof Given $H_k \succ 0$, nonzero g_k gives $q_k, d_k > 0$; weak Wolfe gives $t_k \geq (1 - c_2) d_k > 0$. For any vector v, the quadratic form in (2) is a sum of two nonnegative terms. If both vanish, then $s_k^T v = 0$ and $(I - y_k s_k^T / t_k) v = 0$, whence $v = 0$. Thus $H_{k+1} \succ 0$, proving the induction. Armijo gives the first lower bound in (4); the supporting inequality for F gives the upper bound. Summing the lower bound proves summability when the values have a finite limit.

The Hessian identity is the classical equivalent form of BFGS [7, Section 5]; it also follows by multiplying its right side by (2), using $B_k s_k = -\alpha_k g_k$ and $s_k^T y_k = t_k$. Finally, metric Cauchy–Schwarz gives $(g_k^T v)^2 \leq q_k v^T B_k v$, and $g_k^T s_k = -\alpha_k q_k$ gives $\widehat{B}_k s_k = 0$. □

Lemma 2 (Reference energy) *For any fixed x_*, put $r_k = x_k - x_*$ and*

$$E_k = r_k^T B_k r_k, \qquad a_k = g_k^T r_k, \qquad R_k = \frac{(y_k^T r_{k+1})^2}{t_k}.$$

Then

$$E_{k+1} = E_k - \frac{a_k^2}{q_k} + R_k. \tag{6}$$

If x_* minimizes F, with $F_* = F(x_*)$, then

$$(F(x_N) - F_*)^2 \sum_{k<N} q_k^{-1} \le E_0 + \sum_{k<N} R_k. \tag{7}$$

Proof Since $r_{k+1} = r_k + s_k$ and $\widehat{B}_k s_k = 0$, the Hessian update gives

$$r_{k+1}^T B_{k+1} r_{k+1} = r_k^T \widehat{B}_k r_k + \frac{(y_k^T r_{k+1})^2}{t_k},$$

which is (6). At a minimizer, convexity gives $a_k \ge F(x_k) - F_*$; the values decrease. Summation and $E_N \ge 0$ yield (7). □

The identity makes the intended contradiction visible. If a gap of at least $\eta > 0$ persists and the secant correction satisfies $R_k \le C$, then (7) gives

$$\eta^2 \sum_{k<N} q_k^{-1} \le E_0 + CN.$$

Thus an objective-level bound on one component of a gradient jump would force the reciprocal sum to grow at most linearly. Theorem 2 proves that, along runs with bounded gradients, the full update requires faster growth. Neither step requires a bound on $\|B_k\|$ or a limit for H_k: the initial energy is finite, and the later energies need only remain nonnegative.

3.1 Principal minors and reciprocal growth

For a symmetric $n \times n$ matrix H, define

$$e_j(H) = \sum_{\substack{I \subset \{1,\dots,n\} \\ |I| = j}} \det H_{I,I}, \qquad e_{n+1}(H) = 0.$$

Multilinearity of the determinant identifies these sums with the coefficients of $\det(I + zH)$; hence they are invariant under orthogonal changes of coordinates and equal the elementary symmetric polynomials of the eigenvalues. In particular $e_1 = \mathrm{tr}$ and $e_n = \det$. Trace and determinant potentials are central to classical BFGS analysis [7,8]. The determinant records volume, while the intermediate principal-minor sums retain information about different-dimensional compressions. The inequality below couples each degree to the next one. After normalization, this coupling supplies a positive weighted potential even when the matrices become poorly conditioned.

Theorem 1 (Principal-minor hierarchy) *Let $H \succ 0$, $s = -\alpha H g$, $\alpha > 0$, and suppose*

$$q = g^T H g > 0, \qquad \beta = -\frac{y^T H g}{q} > 0.$$

For the inverse-BFGS update H_+ in (2)*,*

$$e_j(H_+) \le \frac{\|y\|^2}{\beta^2 q} e_{j+1}(H) + \frac{\alpha}{\beta} e_j(H), \qquad 1 \le j \le n. \tag{8}$$

At $j = n$ the exact identity is $\det H_+ = (\alpha/\beta) \det H$.

Proof Let Z have j orthonormal columns. The complete compression calculation in Lemma 6 of Appendix A gives

$$\det(Z^T H_+ Z) \le \frac{\det \mathrm{Gram}_H(y, Z)}{\beta^2 q} + \frac{\alpha}{\beta} \det(Z^T H Z),$$

where $\mathrm{Gram}_H(y, Z) = [y\ Z]^T H [y\ Z]$. Choose an orthonormal eigenbasis of H, with eigenvalues $\lambda_i > 0$, and write y_i for the coordinates of y in this basis. Summing over its j-column selections Z_I gives

$$\begin{aligned} \sum_{|I|=j} \det \mathrm{Gram}_H(y, Z_I) &= \sum_{|I|=j} \left(\prod_{i \in I} \lambda_i \right) \sum_{\ell \notin I} \lambda_\ell y_\ell^2 \\ &= \sum_{|J|=j+1} \left(\prod_{i \in J} \lambda_i \right) \sum_{\ell \in J} y_\ell^2 \le \|y\|^2 e_{j+1}(H). \end{aligned}$$

The first equality is the Schur complement of the diagonal compression; the second groups terms by $J = I \cup \{\ell\}$. Orthogonal invariance now proves (8). The last assertion follows from the full-size case in Lemma 6. □

Theorem 2 (Reciprocal growth) *If an admissible run has bounded gradients and $d_k \to 0$, then*

$$\left(\prod_{k<N} q_k^{-1} \right)^{1/N} \longrightarrow +\infty, \qquad \frac{1}{N} \sum_{k<N} q_k^{-1} \longrightarrow +\infty. \tag{9}$$

Proof Choose $\Lambda > 0$ with $\|g_k\| \le \Lambda$, and put $K_0 = 4\Lambda^2/\omega^2$. For the actual run,

$$\beta_k = -\frac{y_k^T H_k g_k}{q_k} = \frac{t_k}{d_k} \ge \omega, \qquad \|y_k\| \le 2\Lambda, \qquad \alpha_k = \frac{d_k}{q_k}.$$

Theorem 1 therefore yields the normalized hierarchy

$$q_k e_j(H_{k+1}) \le K_0 e_{j+1}(H_k) + \frac{d_k}{\omega} e_j(H_k). \tag{10}$$

This recurrence is derived for the run's principal determinants.

Fix $\eta > 0$ and choose J so that $d_k/\omega \le \eta$ for $k \ge J$. For $H \succ 0$ define

$$V_\eta(H) = \sum_{j=1}^{n} (K_0/\eta)^{j-1} e_j(H) > 0.$$

For $k \ge J$, multiply (10) by these weights and sum. The e_{j+1} terms contribute at most $\eta V_\eta(H_k)$, as do the remaining terms. Also $q_k \le \Lambda^2 \operatorname{tr} H_k \le \Lambda^2 V_\eta(H_k)$. Consequently

$$V_\eta(H_{k+1}) \le \frac{2\eta}{q_k} V_\eta(H_k), \qquad \prod_{k=J}^{J+m-1} q_k^{-1} \ge \frac{1}{\Lambda^2 V_\eta(H_J)(2\eta)^{m-1}}.$$

For the last bound, iterate through $m-1$ updates and apply the bound on q_{J+m-1} at the remaining endpoint. Taking mth roots gives a lower limit of at least $(2\eta)^{-1}$. A fixed prefix contributes a factor tending to one. Since η is arbitrary, the geometric mean diverges; AM–GM gives the arithmetic-mean conclusion. □

The argument extends the product-estimate tradition in BFGS analysis [6, Theorem 3.7], [10, Appendices C–D] to the bounded-gradient, vanishing-decrease regime through the normalized adjacent-degree hierarchy. The assumption $d_k \to 0$ is weaker than the summability supplied by Armijo on a compact sublevel. Finite-horizon bounds and the stronger reciprocal-sum consequence under summability are retained in the companion's Chapter 2; the qualitative conclusion above is sufficient for the article. Neither limit in (9) asserts that $q_k \to 0$ pointwise or that H_k converges.

3.2 Moving comparison directions

The radial direction is effective for a centered homogeneous term, but a curved switching interface may rotate the direction in which its jump cancels. The comparison geometry must then move. Three requirements have distinct roles. *Visibility* keeps the remaining objective gap detectable on a positive band. *Secant control* limits the gradient-jump component seen by the geometry at the next endpoint. *Transport* bounds the cost of changing that geometry between endpoints. The identity $\widehat{B}_k s_k = 0$ means that movement parallel to the executed step carries no cost in the part of the update left after its negative rank-one term. Only the perpendicular projection needs control.

A positive-semidefinite field packages one or several comparison directions. It is constructed from the objective for the analysis; BFGS neither evaluates it nor changes its step in response to it. The rank-one radial field in Section 4 and the contour field in Section 5 illustrate the two designs.

For differentiable x, z, write $g = \nabla F(x)$, $h = \nabla F(z)$, $s = z - x$, $y = h - g$, $d = -g^T s$, $\Delta = F(x) - F(z)$ and $t = y^T s$. Call the pair eligible when

$$d > 0, \qquad c_1 d \le \Delta \le d, \qquad t \ge \omega d.$$

All accepted pairs are eligible. Requiring a property of every eligible pair is an objective assumption; it does not assert that an arbitrary sequence of eligible pairs can be realized by BFGS. Put $\Pi_s = I - ss^T/(s^T s)$.

Definition 2 (Band certificate) Let $K = \{F \le b\}$ be compact, with $b > F_* = \min F$. For each nonempty $K_\eta = \{x \in K : F(x) \ge F_* + \eta\}$, $\eta > 0$, a band certificate consists of a finite-valued symmetric PSD field $W_\eta(x)$ on its differentiable points and constants $m_\eta > 0$, $C_\eta \ge 0$, $L_\eta \ge 0$, $\delta_\eta > 0$ such that

$$g^T W_\eta(x) g \ge m_\eta, \tag{11}$$

$$y^T W_\eta(z) y \le C_\eta t, \tag{12}$$

$$\Pi_s W_\eta(z) \Pi_s \preceq (1 + L_\eta d)\Pi_s W_\eta(x)\Pi_s. \tag{13}$$

The first inequality holds at every differentiable point of K_η; the last two hold for all eligible pairs in K_η with $d \le \delta_\eta$.

Neither continuity, uniform boundedness nor positive definiteness of W_η is required. The constants and field may depend on the positive band. The positive-band restriction permits these constants to deteriorate near the solution set: the proof only needs to rule out a limiting gap bounded away from zero. In particular, the condition allows nonsmoothness at an optimizer.

Theorem 3 (Transported comparison certificate) *If K admits the band certificates above, every admissible run with $x_0 \in K$ satisfies*

$$F(x_k) \to F_*, \qquad \operatorname{dist}(x_k, \arg\min F) \to 0.$$

If the minimizer is unique, the iterates converge to it.

Proof Armijo keeps the run in K, and its values have a finite limit. Lemma 1 gives $\sum d_k < \infty$. Gradients are bounded there: choose $\epsilon > 0$ and bound the continuous finite convex function on $K + \epsilon\overline{B}$. Its supporting inequality at x evaluated at $x + \epsilon u$, for every unit u, bounds every subgradient at x. Thus Theorem 2 applies.

Suppose the limiting gap is $\ell > 0$. On a tail $k \ge J$, use the field on $K_{\ell/2}$ and ensure $d_k \le \delta_{\ell/2}$; omit the band subscripts. Set $E_k = \operatorname{tr}(B_k W(x_k)) \ge 0$. Since $\widehat{B}_k = \Pi_{s_k} \widehat{B}_k \Pi_{s_k} \succeq 0$, trace monotonicity and (11)–(13) give

$$\begin{aligned} E_{k+1} &= \operatorname{tr}(\widehat{B}_k W(x_{k+1})) + \frac{y_k^T W(x_{k+1}) y_k}{t_k} \\ &\le (1 + L d_k) \operatorname{tr}(\widehat{B}_k W(x_k)) + C \le (1 + L d_k)(E_k - m/q_k) + C. \end{aligned}$$

Trace monotonicity here follows from $\operatorname{tr}(AQ) = \operatorname{tr}(Q^{1/2} A Q^{1/2}) \ge 0$ for PSD A, Q. Let $P_J = 1$ and $P_{k+1} = P_k(1 + L d_k)$. Summability gives $1 \le P_k \le \exp(L \sum_{j \ge J} d_j) =: \overline{P}$. Dividing by P_{k+1}, summing and using $E_N \ge 0$ yields

$$m \overline{P}^{-1} \sum_{k=J}^{N-1} q_k^{-1} \le E_J + C(N - J),$$

contradicting (9). This normalization is the usual summable multiplicative-error argument; compare [11, Lemmas 2.2–2.3]. Finally, a sequence in compact K with values tending to F_* cannot stay a positive distance from the closed solution set: a convergent subsequence would have a nonoptimal limit with value F_*. With a unique minimizer the same argument gives convergence of the entire sequence. □

The companion's scalar convergence and certificate-nonexistence results show that the certificate is not necessary for successful BFGS behavior. The planar application below shows a different point: moving geometry can prove convergence where the fixed radial field fails.

4 Global convergence: homogeneous and mixed geometry

Toward a minimizer, a homogeneous term has an exact first-order model; a smooth convex term adds only a quadratic remainder. We turn this inward accuracy into the secant bound needed by the comparison energy, then apply it to centered sums and coercive homogeneous objectives in any finite dimension.

4.1 Why inward accuracy controls a nonradial algorithm

A gradient jump can make the positive correction in (6) large. We control its pairing with the direction to a minimizer by asking that first-order prediction remain quadratically accurate on short inward segments. This asks nothing about transverse smoothness, and it does not require BFGS to take radial steps. The comparison segments are an analytical device; the actual steps remain those in (2).

Fix a minimizer x_*, write $F_* = F(x_*)$, and let $K = \{x : F(x) \le b\}$ be compact, with $b > F_*$. For differentiable $x, z \in K$, set

$$r_x = x - x_*, \qquad g = \nabla F(x), \qquad h = \nabla F(z),$$
$$T = (h-g)^T(z-x), \qquad N = (h-g)^T r_z.$$

Here T is the quantity t of Section 3, and N is the *radial numerator*. The radial secant condition is the following requirement on positive bands:

$$\begin{gathered}\text{for every } \eta > 0 \text{ there is } C_\eta \ge 0 \text{ such that}\\ N^2 \le C_\eta T \quad \text{whenever } F_* + \eta \le F(z) \le F(x), \quad T > 0.\end{gathered} \tag{14}$$

Here all such differentiable endpoint pairs are included, whether or not they are eligible for the fixed line search. The inward model condition is

$$\begin{gathered}\text{for every } \eta > 0 \text{ there are } M_\eta \ge 0,\ \theta_\eta \in (0, 1/2] \text{ such that}\\ 0 \le F(x - \tau r_x) - F(x) + \tau \nabla F(x)^T r_x \le \frac{M_\eta}{2}\tau^2, \quad 0 \le \tau \le \theta_\eta,\end{gathered} \tag{15}$$

at every differentiable $x \in K$ with $F(x) \geq F_* + \eta$. The inward point need not be differentiable. It remains in K by convexity. The parameter τ is a fraction of the distance to x_*, rather than a Euclidean step length. Constants may depend on the positive value band: no common bound is required as the band approaches the minimum. Thus nonsmoothness at the solution is permitted, including the absolute-value term in the benchmark.

Theorem 4 (Inward models control radial secants) *For finite convex F and the compact sublevel K above,* (15) *implies* (14)*.*

Proof Choose $\Lambda, R > 0$ bounding subgradient norms on K and $\|x - x_*\|$ there. The subgradient bound was shown in the proof of Theorem 3. Use the supporting-plane error

$$D_F(z, x) = F(z) - F(x) - \nabla F(x)^T (z - x) \geq 0.$$

This is the usual Bregman expression [4]; only its supporting endpoint needs to be differentiable. At differentiable endpoints, $T = D_F(z, x) + D_F(x, z)$.

Suppose that (15) holds, and fix a positive band, with constants M, θ. If $N \geq 0$, compare the supporting plane at x with the inward model at z, both evaluated at $z - \tau r_z$, to obtain

$$\tau N \leq D_F(z, x) + \tfrac{1}{2} M \tau^2.$$

If $N < 0$, reverse the endpoints and use the inward model at x. The reversed pair has radial numerator $(g - h)^T r_x = T - N \geq |N|$, and

$$\tau(T - N) \leq D_F(x, z) + \tfrac{1}{2} M \tau^2.$$

Thus there are $A \in [|N|, 2\Lambda R]$ and $D \in \{D_F(z, x), D_F(x, z)\}$ such that $\tau A \leq D + M\tau^2/2$ for $0 \leq \tau \leq \theta$. For $M > 0$ and $A \leq M\theta$, choose $\tau = A/M$, giving $A^2 \leq 2MD$. If $A > M\theta$, take $\tau = \theta$ to obtain $D \geq A\theta/2$ and hence $A^2 \leq (4\Lambda R/\theta)D$. When $M = 0$, the choice $\tau = \theta$ gives an even stronger estimate. Therefore

$$N^2 \leq \max\{2M, 4\Lambda R/\theta\}\, T. \qquad \square \tag{16}$$

The converse implication, including inward samples that are not differentiable, is proved in Chapter 4 of the complete companion. The article uses the sufficient direction just proved: objective modeling accuracy controls every descending secant pair on the same positive band.

Corollary 1 (Radial convergence) *Under either condition, every admissible run starting in K converges in value to F_* and in distance to* $\arg\min F$*. With a unique minimizer it converges to x_*.*

Proof Take $W_\eta(x) = r_x r_x^T$. Convexity gives $g^T r_x \geq F(x) - F_* \geq \eta$, proving visibility with $m_\eta = \eta^2$. By Theorem 4 we may assume (14); it gives secant control on eligible pairs in K_η, since these have $F(z) < F(x)$ and $T = t > 0$. Finally $r_z = r_x + s$ implies $\Pi_s r_z = \Pi_s r_x$, so transport holds with $L_\eta = 0$. Apply Theorem 3. $\square$

Condition (15) controls only inward first-order modeling error. In a smooth region its infinitesimal expression is $r_x^T \nabla^2 F(x) r_x$, rather than the largest Hessian eigenvalue. A bound almost everywhere on that expression alone would miss a jump of the derivative along a ray; the function-value condition (15) includes such jumps. The condition is sufficient; Theorem 9 will exhibit convergence where (14), and hence by Theorem 4 also (15), fails on positive bands arbitrarily near the minimizer.

4.2 A family theorem and the mixed benchmark

A finite convex positively homogeneous function is sublinear: its value on $(1-\tau)r$ is exactly $(1-\tau)$ times its value on r. Its angular nonsmoothness consequently disappears from the inward remainder. The next theorem permits an arbitrary locally $C^{1,1}$ convex perturbation, provided the homogeneous part is centered at a minimizer of the sum. Centering is a geometric hypothesis on the objective, not knowledge required by the algorithm. Moreover, the two summands need not separately be minimized at that center.

Theorem 5 (Centered smooth-plus-sublinear objectives) *Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, and suppose x_* minimizes F. Let U be an open convex set, let ψ be convex and locally $C^{1,1}$ on U, and let $\gamma : \mathbb{R}^n \to \mathbb{R}$ be finite, convex and positively homogeneous of degree one. Suppose*

$$F(x) = \psi(x) + \gamma(x - x_*) \qquad (x \in U).$$

Every admissible run whose initial sublevel is compact and contained in U satisfies $F(x_k) \to F(x_*)$. The distance to $\arg\min F$ *also tends to zero. If x_* is the unique minimizer, then $x_k \to x_*$. Neither original summand is required to be minimized at x_*.*

Proof Let $K = \{F \le F(x_0)\}$. Since $g_0 \neq 0$, convexity implies $F(x_0) > F(x_*)$. On compact convex $K \subset U$, local $C^{1,1}$ regularity provides a uniform Lipschitz constant L for $\nabla\psi$: take a finite cover by neighborhoods carrying local constants and subdivide any segment in K into pieces shorter than a Lebesgue number of that cover. Summing the local bounds over the pieces gives the largest of those constants as a Lipschitz constant along the segment. Integrating the gradient along an inward segment consequently gives

$$\left|\psi(x - \tau r_x) - \psi(x) + \tau \nabla\psi(x)^T r_x\right| \le \tfrac{1}{2} L \tau^2 \|r_x\|^2.$$

At a differentiable point of F in U, the representation and smoothness of ψ imply differentiability of γ at r_x. Differentiating its homogeneity identity gives $\nabla\gamma(r_x)^T r_x = \gamma(r_x)$. Its contribution to the inward remainder is therefore exactly zero. The upper bound in (15) follows with $M = L \max_{x \in K} \|r_x\|^2$ and $\theta = 1/2$ for every band; its lower bound follows from convexity of F. Corollary 1 proves the claim. □

The compact initial sublevel supplies uniform constants and traps the accepted iterates. When the representation is global and all sublevels are compact, the theorem applies from every differentiable nonstationary starting point, for every initial SPD matrix and every fixed valid Armijo–weak-Wolfe pair. Neither a lower Hessian bound for the smooth term nor differentiability of the full objective away from the minimizer is required. Without uniqueness, the stated conclusion is value and solution-set distance convergence; convergence to one particular minimizer is not asserted.

Corollary 2 (The quadratic–absolute-value benchmark) *For $F(u, v) = u^2 + |v|$, every infinite admissible full-BFGS run, with arbitrary $H_0 \succ 0$ and any fixed $0 < c_1 < c_2 < 1$, satisfies*

$$F(u_k, v_k) \longrightarrow 0, \qquad (u_k, v_k) \longrightarrow (0, 0).$$

Proof Take $\psi(u, v) = u^2$, $\gamma(u, v) = |v|$, $x_* = (0, 0)$ and $U = \mathbb{R}^2$. The objective is finite and convex, its sublevels are closed and bounded, and its minimizer is unique. Equation (1) shows the exact inward error. Theorem 5 applies. □

This strengthens Guo–Lewis's subsequence approach to the line $v = 0$ [5, Corollary 4.4] to the whole-sequence convergence to the origin they left open, under the admissible full-BFGS convention of this paper.

4.3 Coercive homogeneous objectives, linear maxima, and gauges

Pure homogeneous geometry is the limiting case with no smooth remainder. Coercivity here has a clear purpose: it prevents escape along a direction of zero or negative homogeneous value and converts convergence in value into convergence to the origin. It imposes no smoothness or finite-piece condition.

Theorem 6 (Coercive homogeneous objectives) *Let $F : \mathbb{R}^n \to \mathbb{R}$, $n \geq 1$, be finite, convex, positively homogeneous of degree one, and coercive. Then zero is its unique minimizer and every infinite admissible full-BFGS run satisfies*

$$F(x_k) \longrightarrow 0, \qquad x_k \longrightarrow 0.$$

The initial point, initial SPD matrix and fixed valid line-search constants are arbitrary within admissibility.

Proof Homogeneity gives $F(0) = 0$. If $F(x) \leq 0$ for some $x \neq 0$, then $F(tx) = tF(x) \leq 0$ as $t \to \infty$, contradicting coercivity. Thus $F(x) > 0$ off zero. Continuity on the unit sphere gives

$$\mu = \min_{\|x\|=1} F(x) > 0, \qquad F(x) \geq \mu \|x\|.$$

All sublevels are compact. Theorem 5, with $x_* = 0$, $U = \mathbb{R}^n$, $\psi = 0$ and $\gamma = F$, gives convergence in value, and the displayed lower bound gives convergence of the iterates. Alternatively, the inward remainder is identically zero, so the radial convergence corollary applies directly. □

Corollary 3 (Compact homogeneous models) *The conclusion of Theorem 6 holds for every norm, every finite convex degree-one gauge with bounded unit sublevel, and every finite linear maximum*

$$F(x) = \max_{1 \le i \le m} a_i^T x$$

whose zero sublevel is bounded. These are whole-iterate conclusions from every admissible start. The linear maximum has no affine intercepts.

Proof For a norm, continuity and positivity give a positive minimum on the Euclidean unit sphere. For either of the other models, boundedness of the stated sublevel rules out a nonzero x with $F(x) \le 0$, since its full positive ray would lie in that sublevel. The same sphere argument gives $F(x) \ge \mu\|x\|$, hence coercivity. Finite maxima of linear forms are finite, convex and degree-one homogeneous, so the theorem applies. □

The norm cases include $\|x\|_1 = \sum_i |x_i|$ and $\|x\|_\infty = \max_i |x_i|$. In dimension at least two their nonsmoothness is nonisolated: the first has gradient jumps across coordinate hyperplanes; the second has them where two nonzero coordinate magnitudes tie for the maximum. For example, along $(t, t, 0, \ldots, 0)$ with $t > 0$, the two active coordinate forms of $\|\cdot\|_\infty$ have distinct gradients. These are concrete specializations of the norm corollary, with no separate priority claim. The familiar Euclidean norm is an already established consequence of Guo–Lewis's smooth-extension analysis [5, Corollary 4.2]; the scalar absolute-value model was analyzed by Lewis–Overton under their stated search conventions [1, Section 5.1]. No symmetry is needed for the gauge or max-linear cases. Coercivity supplies this compact, whole-iterate route. The complete companion, Chapter 7, separately proves the noncompact homogeneous value dichotomy $F(x_k) \to 0$ or $F(x_k) \to -\infty$, together with its bounded-below and additional iterate consequences.

5 Local convergence: geometry that survives near a minimizer

Theorem 5 requires the initial sublevel to lie in the region where its representation is valid. A local theorem must choose that region before seeing a BFGS run or its initial metric. Convexity and a unique minimizer supply precisely this localization. We first use it for finite polyhedral pieces, where inward accuracy has a direct geometric explanation, and then for curved planar switching interfaces, where a moving comparison direction is essential to the proof.

5.1 Localization and finite polyhedral pieces

The distinction between a local objective hypothesis and a hypothesis on the whole initial sublevel is handled by a simple convex trapping argument.

Lemma 3 (A common local basin) *Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex with a unique minimizer x_*. Every sublevel is compact. For each neighborhood U of x_*, there is $\varepsilon > 0$ such that $\{F \le F(x_*) + \varepsilon\} \subset U$. Thus any admissible run starting in that sublevel stays in U, independently of its initial SPD matrix.*

Proof Choose $r > 0$ with $\overline{B}_r(x_*) \subset U$. Continuity, compactness of the sphere and uniqueness give

$$\delta = \min_{\|u\|=r} \big(F(x_* + u) - F(x_*)\big) > 0.$$

For $\|x - x_*\| \ge r$, convexity at the point at distance r on the segment from x_* to x gives

$$F(x) - F(x_*) \ge \frac{\|x - x_*\|}{r}\delta.$$

Consequently F is coercive, hence has compact sublevels, and $\varepsilon = \delta/2$ gives the asserted inclusion. Armijo preserves it. □

A finite local partition controls which formulas an inward comparison segment can cross. Near x_* only cells containing x_* occur: the finitely many closed cells that miss x_* avoid a ball around it. Convexity of a cell then keeps the entire inward segment in that same cell. Its smooth extension yields quadratic model accuracy even if the full gradient jumps across cell boundaries. The algorithm need not identify a cell or a manifold; only the comparison segment uses the cell geometry.

Theorem 7 (Local convergence on finite polyhedral cells) *Let $n \ge 1$ and let $F : \mathbb{R}^n \to \mathbb{R}$ be finite and convex, with a unique minimizer x_*. Suppose an open neighborhood U of x_* is covered by finitely many closed full-dimensional polyhedra P_i. For each i, suppose there is an open set $V_i \supset U \cap P_i$ and a function $f_i \in C^{1,1}_{\mathrm{loc}}(V_i)$ such that*

$$F(x) = f_i(x) \qquad (x \in U \cap P_i).$$

Then there exists $\varepsilon > 0$ such that every infinite full-BFGS run with arbitrary $H_0 \succ 0$, differentiable accepted iterates, fixed Armijo–weak-Wolfe constants $0 < c_1 < c_2 < 1$, and $F(x_0) \le F(x_*) + \varepsilon$ satisfies

$$x_k \in U \quad (k \ge 0), \qquad F(x_k) \to F(x_*), \qquad x_k \to x_*.$$

The basin is chosen before the run and its initial matrix. Only the full objective must be convex; the extensions need not be convex outside their cells.

Proof (Theorem 7) First discard cells not containing x_*. Each such closed cell has an open complement containing x_*; finiteness supplies one ball that avoids them all. For every remaining cell, its extension domain V_i contains x_*. Shrink to a common ball $B_\rho(x_*) \subset U \cap \bigcap_{i : x_* \in P_i} V_i$, and then shrink again so that the gradients of all remaining extensions have one Lipschitz bound L there. This uses only finitely many local extensions; they need not have shared a domain before shrinking.

Let x be a differentiability point of F in this ball, and choose a cell P_i containing it. Convexity of P_i places the whole inward segment $x - \tau(x - x_*)$, $0 \le \tau \le 1$, in P_i. Both F and f_i agree along that segment. Comparing their right derivatives at $\tau = 0$ gives

$$\nabla F(x)^T(x - x_*) = \nabla f_i(x)^T(x - x_*).$$

Only equality in the radial direction is needed, including at cell boundaries. The smooth Taylor estimate along the same segment gives

$$0 \le F(x - \tau r_x) - F(x) + \tau \nabla F(x)^T r_x \le \tfrac{1}{2} L\tau^2 \|r_x\|^2 \le \tfrac{1}{2} L\rho^2\tau^2.$$

The first inequality uses convexity of F, not convexity of the extension. Lemma 3 supplies a compact positive sublevel wholly contained in this ball, chosen before the run or its matrix. The inward condition holds on that sublevel, with $M = L\rho^2$ and $\theta = 1/2$ for every band, so Corollary 1 gives value and iterate convergence. Armijo gives the asserted containment. □

Corollary 4 (Smooth losses with polyhedral or quadratic pieces) *A finite convex piecewise linear-quadratic objective with a unique minimizer has the local convergence property of Theorem 7. The same holds for a convex locally $C^{1,1}$ loss plus a finite convex piecewise linear-quadratic regularizer, provided the sum has a unique minimizer.*

Proof By piecewise linear-quadratic we mean a representation by quadratic polynomials on a finite cover by closed full-dimensional polyhedral cells. Each polynomial is a smooth extension of its cell formula; adding the smooth loss preserves the required local $C^{1,1}$ extensions. Theorem 7 applies. □

In particular, a finite convex polyhedral penalty, such as a finite sum of absolute-value or hinge terms, is piecewise affine on a finite polyhedral partition. It therefore falls under this corollary when added to a locally $C^{1,1}$ convex loss, under the stated uniqueness assumption. Different cells may have different quadratic Hessians. A finite maximum of arbitrary quadratic formulas need not admit such a polyhedral partition and is not automatically covered. There is also no claim that iterates identify a cell or that rejected trials remain in the neighborhood U of Theorem 7.

Corollary 5 (A local centered-composite basin) *If the representation and smoothness assumptions of Theorem 5 hold on a neighborhood of a unique minimizer x_*, then some $\varepsilon > 0$, chosen from the objective alone, has the following property: every admissible run starting with $F(x_0) \le F(x_*) + \varepsilon$ stays in that neighborhood and converges to x_* in value and iterates.*

Proof Shrink to an open convex neighborhood where the representation holds. Lemma 3 gives a compact sublevel contained there. Armijo traps the accepted points, and Theorem 5 applies. □

For a concrete reduction, let $F(x) = \frac{1}{2}\|Ax - b\|^2 + \lambda\|x\|_1$, with $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $\lambda > 0$, and assume that F has a unique minimizer x_*. Put $I_0 = \{i : x_{*,i} = 0\}$ and $\sigma_i = \operatorname{sign}(x_{*,i})$ for $i \notin I_0$. On the open convex neighborhood where $\sigma_i x_i > 0$ for all $i \notin I_0$, the exact decomposition is

$$\begin{aligned} F(x) &= \psi(x) + \gamma(x - x_*), \\ \psi(x) &= \tfrac{1}{2}\|Ax - b\|^2 + \lambda \sum_{i\notin I_0} \sigma_i x_i, \\ \gamma(r) &= \lambda \sum_{i\in I_0} |r_i|. \end{aligned} \tag{17}$$

The nonzero optimal coordinates contribute affine terms to the smooth convex function ψ; the zero coordinates give the centered sublinear function γ. Corollary 5 therefore gives a positive sublevel basin, chosen independently of the initial SPD matrix, in which every admissible run converges in value and iterates to x_*. Uniqueness is a hypothesis here; $\lambda > 0$ alone does not supply it.

A locally $C^{1,1}$ convex loss plus finitely many Euclidean residual norms $\|A_j x - b_j\|$ has this local form. A residual nonzero at x_* remains nonzero nearby, so its Euclidean norm is smooth there; a zero residual gives the centered sublinear function $\|A_j(x - x_*)\|$. Summing these decompositions proves the representation. Under uniqueness, the corollary applies. This example allows nonpolyhedral penalties while retaining the same radial mechanism.

5.2 A moving direction for curved planar switching

The radial direction can fail for a different reason: a curved interface may carry a gradient jump with a nonzero radial component. The secant correction then blows up as endpoints approach opposite sides of the interface. The transported framework allows the comparison direction to follow the interface instead. It must first remain visible to every local supporting model, then cancel the leading jump, and finally move at a cost controlled by the actual objective decrease. The field is constructed from the objective; BFGS never evaluates it.

For $F : \mathbb{R}^2 \to \mathbb{R}$, write

$$\mathcal{C}_{\text{aff}} = \{p : 0 \in \operatorname{aff} \partial F(p)\}.$$

At a point outside this set, let a be the least-norm point of $\operatorname{aff} \partial F(p)$. Orthogonality to its direction space gives $g^T a = \|a\|^2$ for every $g \in \partial F(p)$. Thus $-a/\|a\|^2$ decreases all supporting models at the same unit rate. Affine criticality concerns failure of this common-rate direction; ordinary stationarity concerns failure of every strict common descent direction. They need not coincide.

The two planar hypotheses have separate roles. Strictly convex positive sublevels prohibit long accepted chords with vanishing value decrease. Regular two-branch switching makes each positive-level corner move smoothly with the level and supplies a direction paired equally with both gradients. Neither condition imposes a matrix limit or a lower bound on all eigenvalues of a

Hessian. A common positive lower Hessian bound is one sufficient way to obtain strict contours, but is not required.

Here is the construction near a compact non-affine-critical part of a positive level. At a two-branch corner, the independent gradients let us solve $f_1(p_j(t)) = f_2(p_j(t)) = t$ for the corner as the level varies. Its velocity $v_j(t) = p_j'(t)$ satisfies $\nabla f_i(p_j(t))^T v_j(t) = 1$ for both branches. Thus $-v_j$ is an equal-rate descent direction, while v_j is the positively paired field used in the energy. There are finitely many such corners in the compact part under consideration. Their supporting-normal arcs are disjoint: a common normal at two corners would expose the whole segment between them, contrary to the strict-contour assumption. Their angular separation leaves room to join the local velocities smoothly.

The joining also has to control derivatives. For unit normal $n(\theta) = (\cos\theta, \sin\theta)$ and tangent $T(\theta) = (-\sin\theta, \cos\theta)$, choose a positive angular function $h(t, \theta)$ that equals $n(\theta)^T v_j(t)$ near the jth normal arc. Disjoint smooth cutoffs blend these positive pairings with $h = 1$ elsewhere. The field $w = hn + h_\theta T$, evaluated at $t = F(x)$ and the gradient's normal angle, equals $v_j(F(x))$ on both sides of that corner. Its angular derivative is $(h + h_{\theta\theta})T$: the normal terms cancel. Consequently Dw preserves the tangent line of each smooth level. The one-dimensional tangent is essential here: for a tangent step s, this makes $Dw\, s$ parallel to s, so $\det(s, Dw\, s) = 0$.

For a sufficiently short eligible chord, weak Wolfe makes $\det(s, w)$ uniformly nonzero relative to $\|s\|$. Tangent preservation then bounds the change of $\log|\det(s, w)|$ by a constant times the variation of F along the chord. Convexity bounds that variation by $2d$, giving $|\log\lambda| \le Cd$ for the transport factor λ in $w(z) = \lambda w(x) + \beta s$. Since Armijo gives $d \le (F(x) - F(z))/c_1$, transport is charged to summable objective decreases; finite total path length is not required. Appendix B supplies the cutoffs, estimates through switching points, and secant bound. These yield one field on a neighborhood of that compact part of the level, not just separate directions at its corners.

Theorem 8 (An affine-critical cluster point in the plane) *Let $F : \mathbb{R}^2 \to \mathbb{R}$ be finite and convex, attain its minimum F_*, and have a compact sublevel $K = \{F \le b\}$ with $b > F_*$. Assume the following objective geometry:*

1. *For every $F_* < t \le b$, the boundary of $\{F \le t\}$ contains no nontrivial line segment.*
2. *At every $p \in K \setminus \mathcal{C}_{\mathrm{aff}}$, either F is C^2 in a neighborhood of p, or it equals $\max\{f_1, f_2\}$ there, where f_1, f_2 are convex C^2 functions, both active at p, and $\nabla f_1(p), \nabla f_2(p)$ are linearly independent.*

Every infinite admissible full-BFGS run starting in K has an affine-critical cluster point. No local representation is required at affine-critical points.

Proof The values decrease to $\ell \ge F_*$, the gradients are bounded on K, and $\sum d_k < \infty$. Let Γ be the nonempty compact cluster set. Every point of Γ has value ℓ. If $\ell = F_*$, its points are minimizers and hence affine critical. Suppose, for a contradiction, that Γ contains no affine-critical point. Then $\ell > F_*$.

The steps tend to zero. Otherwise compactness gives a subsequence with $x_k \to x$, $x_{k+1} \to z \neq x$. For $0 \leq \theta \leq 1$, the supporting inequality and convexity give

$$F(x_k) - \theta d_k \leq F((1-\theta)x_k + \theta x_{k+1}) \leq (1-\theta)F(x_k) + \theta F(x_{k+1}).$$

Passing to the limit shows $F = \ell$ on $[x, z]$. Such level points lie on the boundary of $\{F \leq \ell\}$: if one were interior, extending the line from a minimizer slightly beyond that point and applying convexity would give value strictly less than ℓ. The resulting boundary segment contradicts the first hypothesis.

Lemma 7, proved in Appendix B, constructs a continuous comparison field w on a neighborhood of the whole set Γ. For each sufficiently close eligible pair in that neighborhood with $d \leq 1$, it gives uniform constants $m > 0, C, L \geq 0$ and scalars $\lambda > 0, \beta$ with

$$\begin{aligned} &(g^T w(x))^2 \geq m, \qquad (y^T w(z))^2/t \leq C, \\ &w(z) = \lambda w(x) + \beta s, \qquad \lambda^2 \leq 1 + Ld. \end{aligned}$$

Compactness implies $\operatorname{dist}(x_k, \Gamma) \to 0$; with $s_k \to 0$ and $d_k \to 0$, these estimates apply to the entire tail, not just a selected subsequence. With $E_k = w(x_k)^T B_k w(x_k)$, the Hessian update and $\widehat{B}_k s_k = 0$ yield

$$E_{k+1} = \lambda_k^2 \left(E_k - \frac{(g_k^T w(x_k))^2}{q_k} \right) + \frac{(y_k^T w(x_{k+1}))^2}{t_k} \leq (1 + Ld_k)(E_k - m/q_k) + C.$$

The parenthesized quadratic form is nonnegative. Dividing by $\prod_{j=J}^{k}(1 + Ld_j)$ and summing, exactly as in the proof of Theorem 3, bounds $\sum_{k=J}^{N-1} q_k^{-1}$ by a constant times $N - J + 1$. This contradicts Theorem 2. □

Corollary 6 (When affine criticality is correct) *Under the hypotheses of Theorem 8, assume in addition that* $\mathcal{C}_{\text{aff}} \cap K \subseteq \arg\min F$. *Every admissible run starting in* K *then converges in value to* F_* *and in distance to* $\arg\min F$. *A unique minimizer gives convergence of the entire sequence.*

Proof The theorem supplies an affine-critical cluster point, which is optimal by hypothesis. Its value equals the limiting value of the decreasing objective sequence. Hence that value is F_*, and compactness gives the distance and uniqueness conclusions as in Theorem 3. □

This condition says that nonoptimal points retain equal-rate descent, which is stronger than ordinary descent. The strict-contour and regular-switching assumptions remain essential to this sufficient planar route; it does not give a criterion for every convergent run.

5.3 Nondegenerate smooth maxima and an explicit nonradial basin

The extra affine-correctness condition can be verified from the active models at the minimizer. Positive optimal multipliers ensure that an affine balance cannot lose a branch nearby without losing stationarity; affine independence makes that balance unique and continuous. In the plane these same assumptions leave at most two active branches at a non-affine-critical point, which supplies the required regular switching.

Corollary 7 (Nondegenerate planar smooth maxima) *Let $F : \mathbb{R}^2 \to \mathbb{R}$ be finite and convex with a unique minimizer x_*. Near x_* suppose $F = \max_{1\le i\le m} f_i$, where the branches are convex and C^2. Suppose sufficiently low positive sublevels are strictly convex in the boundary-segment sense of Theorem 8. For the active set I_* at x_*, suppose the gradients are affinely independent and admit weights*

$$\lambda_i^* > 0, \qquad \sum_{i\in I_*} \lambda_i^* = 1, \qquad \sum_{i\in I_*} \lambda_i^* \nabla f_i(x_*) = 0.$$

Then some $\varepsilon > 0$, chosen independently of H_0 and the valid line-search constants, satisfies

$$F(x_0) < F_* + \varepsilon \quad \Longrightarrow \quad F(x_k) \to F_*, \qquad x_k \to x_*$$

for every infinite admissible full-BFGS run. A common positive lower Hessian bound on the locally relevant branches suffices for the contour hypothesis.

Proof Discard the branches inactive at x_* by their strict value gaps. Let $A(x)$ have columns $(\nabla f_i(x), 1)$ for $i \in I_*$. Affine independence makes $A(x_*)$ full column rank, and this persists nearby. If $A(p)\lambda = (0, 0, 1)^T$ is solvable, its unique solution is

$$\lambda(p) = (A(p)^T A(p))^{-1} A(p)^T (0, 0, 1)^T.$$

This expression is continuous and equals λ^* at x_*. Shrink the neighborhood until all its coordinates are positive. The subdifferential of a finite smooth maximum is the convex hull of its active gradients. Indeed, differentiating the finite maximum in a direction gives the maximum of the active gradient pairings. Every convex combination of those gradients supports the objective locally; convexity extends the supporting inequality globally by taking a short initial segment toward any proposed violating point. Conversely, a subgradient outside their convex hull could be strictly separated from that hull by a direction, contradicting the directional-derivative inequality for a subgradient. An affine representation of zero using gradients active at p, extended by zero on inactive indices, must equal $\lambda(p)$. It therefore uses every index and is a convex representation of zero. Thus every nearby affine-critical point is a minimizer.

At most three planar gradients are affinely independent. Away from affine criticality, three cannot be active, since their affine hull would be all of $\mathbb{R}^2$. One active branch gives a smooth chart. For two active branches their gradients are

distinct; if they were linearly dependent, their affine line would contain zero. Thus the gradients are independent, and the other branches have strict gaps. This gives the required convex two-branch chart.

Lemma 3 now puts a sufficiently low compact sublevel inside the chosen neighborhood, with its upper level also below the threshold for strict contours. Apply Corollary 6. If all relevant branch Hessians have a common positive lower bound on a convex neighborhood, subtracting the corresponding positive quadratic makes every branch convex. Their maximum is therefore strongly convex there. A sufficiently low sublevel lies in that neighborhood and cannot have a nontrivial boundary segment, proving the last assertion. □

The multiplier geometry is related to the nondegeneracy used in local bundle Newton analysis (Lewis and Wylie, 2019 preprint, arXiv:1907.11742v1, Propositions 5.5–5.6) and in Gebken's criticality results [3]. Here the full-BFGS conclusion is derived without assumed iterate or matrix asymptotics.

Theorem 9 (Convergence with curved, nonradial switching) *For*

$$\begin{aligned} f_0(u,v) &= u + u^2 + \tfrac{1}{2}v^2, \\ f_1(u,v) &= v + \tfrac{1}{2}u^2 + \tfrac{3}{2}v^2, \\ f_2(u,v) &= -u - v + \tfrac{3}{2}u^2 + v^2, \qquad F = \max_i f_i, \end{aligned}$$

every infinite admissible full-BFGS run with $F(x_0) < 1/100$ converges in value and iterates to zero, for arbitrary $H_0 \succ 0$ and fixed valid Armijo–weak-Wolfe constants. Nevertheless, the radial secant condition (14) *fails on a positive band in every neighborhood of zero. The obstructing endpoint pairs can themselves be chosen eligible for $c_1 = 1/4, c_2 = 1/2$.*

Proof The branch Hessians lie between I and $3I$, and $(f_0+f_1+f_2)/3 = u^2+v^2$. Thus $F \geq u^2 + v^2$, zero is the unique minimizer, and all positive contours are strict. The active gradients at zero are $(1,0), (0,1), (-1,-1)$, with positive weights $1/3$. To obtain the stated basin rather than an unspecified one, put $D = 3(1 + v - uv)$, the determinant of the augmented gradient matrix. Direct solution of that system gives

$$\begin{aligned} D\lambda_0 &= 1 - 4u + 3v - 7uv, \quad D\lambda_1 = 1 + 2u - 3v - uv, \\ D\lambda_2 &= 1 + 2u + 3v + 5uv. \end{aligned}$$

On $|u|, |v| \leq 1/10$, $D \geq 267/100$ and the three numerators are at least $23/100$, $49/100$, and $45/100$. The coefficients sum to one and their weighted gradient is zero. Thus affine independence and affine correctness hold throughout this box, by the same unique-coefficient argument as above. The planar regularity argument also applies throughout the box. Since $\{F \leq 1/100\}$ lies there, Corollary 6 proves the exact convergence claim.

For the radial obstruction choose a sufficiently small fixed $u > 0$ and put

$$v = v(u) = \frac{-1 + \sqrt{1 + 4u + 2u^2}}{2} = u - \tfrac{1}{2}u^2 + O(u^3).$$

This is the positive active tie $f_0 = f_1$, with f_2 inactive. At that point

$$(\nabla f_0 - \nabla f_1)^T(u, v) = (1 + u)u - (1 + 2v)v = \tfrac{1}{2}u^2 - v^2 \neq 0$$

for small u, as follows from the displayed expansion. Take $x = (u, v + e)$ and $z = (u, v - e)$ with $e > 0$ tending to zero. Branches 1 and 0, respectively, are uniquely active. A direct calculation gives

$$\begin{aligned} d &= 2e(1 + 3v + 3e), \quad t = 2e(1 + 2v + 4e), \\ \Delta &= e(1 + 4v) + e^2. \end{aligned}$$

As $u, e \to 0^+$, $\Delta/d \to 1/2$ and $t/d \to 1$, so for fixed sufficiently small u and all sufficiently small e, the pairs are eligible for the stated constants. Their radial numerator $y^T z$ tends to $u^2/2 - v^2 \neq 0$ while $t \to 0^+$. The quotient $(y^T z)^2/t$ is unbounded. For each chosen u both endpoint values stay above half the positive tie value; hence one fixed positive band precedes the unbounded sequence. Choosing u arbitrarily small places that band and those pairs in any given neighborhood of zero. □

The example identifies the extra freedom supplied by moving geometry. Curvature of the interface produces a radial jump, while the contour-based field cancels it and charges its own motion to the accepted decrease. The proof uses the original coordinates and the unmodified BFGS update; a nonlinear change of variables or a replacement by polyhedral cells would not establish this conclusion.

The infinite-run quantifier here has actual witnesses for every initial positive-definite matrix and every valid line-search pair. In particular, $x_0 = (1/1000, 0)$ has $F(x_0) = 1001/10^6 < 1/100$ and can initiate such a run. Corollary 9 verifies existence using differentiability in the full space, in addition to the geometric assumptions checked above. Proposition 4 gives a simpler two-branch family for which those geometric conditions hold on every positive sublevel.

6 Failure: a contracting mechanism and local realizations

The positive theorems identify geometry that permits convergence. We now show why convexity and regularity of the branch formulas cannot supply an unrestricted deterministic guarantee. The first tempting assertion is global: compact sublevels might seem enough to prevent convergence to a wrong corner. The second is local: for one finite maximum of very smooth convex branches, perhaps every sufficiently close start succeeds. We construct an actual failure of each assertion, and then examine what changes when positive lower curvature is imposed.

The mechanism has three parts. A three-step BFGS cycle returns the inverse matrix to a smaller multiple of itself while approaching a nonstationary corner. Strict function and gradient inequalities force the specified line search to produce that cycle. Finally, local copies of the corner accumulate at the true

minimizer of one objective. Matching spatial and objective scales keeps the initial matrix fixed across the copies. The distinction between a formal cycle and an implemented run matters at every part of this argument.

6.1 The specified implementation and the local guarantee at issue

At a differentiable point x, let $g = \nabla F(x)$ and $p = -Hg$. The line search starts with lower bracket $\alpha_{\mathrm{lo}} = 0$, upper bracket $\alpha_{\mathrm{hi}} = +\infty$, and trial length $\alpha = 1$. It tests Armijo first and weak Wolfe second:

$$F(x + \alpha p) \le F(x) + c_1 \alpha g^T p, \qquad \nabla F(x + \alpha p)^T p \ge c_2 g^T p. \tag{18}$$

An Armijo rejection sets $\alpha_{\mathrm{hi}} = \alpha$; a Wolfe rejection sets $\alpha_{\mathrm{lo}} = \alpha$. The next length is $(\alpha_{\mathrm{lo}} + \alpha_{\mathrm{hi}})/2$ when $\alpha_{\mathrm{hi}} < \infty$, and $2\alpha_{\mathrm{lo}}$ otherwise. If both tests hold, the length is accepted. This is the unit-first bracketing implementation of Lewis–Overton [1, Algorithm 4.6]. All comparisons in the construction are strict, so strict acceptance tests give the same decisions. Every queried point in the construction is differentiable, so the Wolfe test is always defined.

Write the full inverse update as

$$\mathcal{B}(H, g, g^+, \alpha) = \left(I - \frac{sy^T}{y^T s}\right) H \left(I - \frac{ys^T}{y^T s}\right) + \frac{ss^T}{y^T s}, \quad s = -\alpha H g, \quad y = g^+ - g. \tag{19}$$

No resetting or rescaling of the stored matrix is performed.

For this routine, even identity initialization does not give a neighborhood in which every differentiable start succeeds under the following regularity assumptions.

Theorem 10 (Local failure with globally smooth convex branches) *There are four globally convex functions $f_i \in C^{2,1}(\mathbb{R}^2)$ such that*

$$0 \preceq \nabla^2 f_i(x) \preceq 64 I \qquad (i = 0, 1, 2, 3; x \in \mathbb{R}^2)$$

and $F = \max_i f_i$ has compact sublevel sets and a unique nonsmooth minimizer at zero. For this one objective there are fixed starting points $x_0^{(\ell)} \to 0$, all with $H_0 = I$, such that, for every fixed pair

$$0 < c_1 \le \tfrac{1}{3}, \qquad c_1 < c_2 < 1,$$

the deterministic unit-first bracketing implementation of Section 6 produces infinite full-BFGS runs satisfying

$$x_k^{(\ell)} \to \bar{x}_\ell \neq 0, \qquad F(x_k^{(\ell)}) \downarrow F(\bar{x}_\ell) > F(0) = 0, \qquad 0 \notin \partial F(\bar{x}_\ell).$$

Every queried point is differentiable. Each line search takes at most five trials, and its accepted lengths repeat $1, 1/4, 1/16$. The objective, initial points, initial matrix and complete decision tree are the same throughout the displayed parameter range. The supremum of the norms of all accepted and rejected query points in the ℓth run tends to zero as $\ell \to \infty$.

The objective and all starts are chosen before the displayed line-search parameters. The failure set supplied by the theorem is a sequence of starts; it is not a positive-measure basin. There is no uniform positive lower curvature requirement on these branches.

6.2 Designing a cycle around a wrong corner

We seek a corner with a strict descent direction that the cycle nevertheless fails to follow. Three slopes in the plane provide the needed freedom. Begin with $H_0 = I$ and fix the first two slopes so that the unit step has a simple exact update. The third slope has two free coordinates. We choose them to close the matrix cycle at a scalar multiple ρI and choose the starting point to close the displacement cycle at ρz_0. For the maximum of linear functions, those two identities allow the same geometry to repeat at each smaller scale.

The target lengths $1, 1/4, 1/16$ fit the bisections of the unit-first search. Choosing them in the algebra does not prove that the algorithm accepts them: we must also exclude every earlier candidate and verify both tests at the intended endpoint. The strict trial certificate below supplies exactly this missing dynamical condition.

Let $(c_\circ, d_\circ)$ be the algebraic pair specified in Appendix C. Define

$$g_0 = (1,0)^T, \qquad g_1 = (-1,-1/8)^T, \qquad g_2 = (c_\circ, d_\circ)^T, \quad G(z) = \max_{0\le i\le 2} g_i^T z.$$

The useful coarse bounds are $0.318 < c_\circ < 0.319$ and $-1.026 < d_\circ < -1.025$. Put

$$\rho = \frac{129(c_\circ - 8d_\circ)}{(16c_\circ - 127d_\circ - 16)(127c_\circ + 16d_\circ + 129)}, \qquad 0.060 < \rho < 0.061. \tag{20}$$

Set

$$M_0 = I, \quad M_1 = \begin{pmatrix} 129/256 & -1/16 \\ -1/16 & 1 \end{pmatrix}, \qquad M_2 = \mathcal{B}(M_1, g_1, g_2, 1/4), \tag{21}$$

$$s_0 = (-1,0)^T, \quad s_1 = (127/1024, 1/64)^T, \qquad s_2 = \rho(1 - c_\circ, -d_\circ)^T. \tag{22}$$

The point is then determined by the displacement closure $s_0 + s_1 + s_2 = -(1-\rho)z_0$; explicitly, let

$$z_0 = \frac{1}{1-\rho}\begin{pmatrix} 897/1024 - \rho(1 - c_\circ) \\ \rho d_\circ - 1/64 \end{pmatrix}, \qquad z_{j+1} = z_j + s_j \quad (j = 0,1,2). \tag{23}$$

Lemma 4 (Exact cycle and strict trial decisions) *The matrices in* (21) *are positive definite, and*

$$\mathcal{B}(I, g_0, g_1, 1) = M_1, \qquad \mathcal{B}(M_2, g_2, g_0, 1/16) = \rho I, \qquad z_3 = \rho z_0.$$

Phase j	$10^4 G$ current	Trial α	$10^4 G$ trial	Active branch	Decision
0	$(8884, 8885)$	1	$(1218, 1219)$	1	accept
1	$(1218, 1219)$	1	$(3845, 3846)$	0	reject
		1/2	$(1364, 1365)$	0	reject
		1/4	$(723, 724)$	2	accept
2	$(723, 724)$	1	$(6687, 6691)$	0	reject
		1/2	$(3406, 3408)$	0	reject
		1/4	$(1765, 1767)$	0	reject
		1/8	$(944, 946)$	0	reject
		1/16	$(534, 536)$	0	accept

Table 2 The deterministic trial tree, read against each phase's current value. "Current" means $G(z_j)$ and "trial" means $G(z_j + \alpha p_j)$. Every rejected trial raises the objective and therefore fails Armijo. The accepted rows also satisfy the strict Armijo and weak-Wolfe margins in (24). The decisions are those of the specified search for every $0 < c_1 \le 1/3$ and $c_2 > 0$. All displayed intervals are rational enclosures, not rounded values.

At z_j, $j = 0, 1, 2$, the unique active gradient is g_j. With $p_j = -M_j g_j$, for every $0 < c_1 \le 1/3$ and $c_2 > 0$ the entire line-search query list is Table 2. At every rejection the objective strictly exceeds its current value. At every acceptance,

$$G(z_j) + \tfrac{1}{3} g_j^T s_j - G(z_{j+1}) > 0, \qquad g_{j+1}^T s_j > 0 > g_j^T s_j, \tag{24}$$

where $g_3 = g_0$. All listed queries and the three base points z_0, z_1, z_2 have a unique active branch, and each such point z, with second coordinate $[z]_2$, satisfies

$$|g_i^T z| < 3 \quad (i = 0, 1, 2), \qquad [z]_2 < 2, \qquad G(z) > 0. \tag{25}$$

The proof of the lemma, including the root isolation, matrix closure and rational certificate, is in Appendix C. The logic of the search can already be read from its margins. Every rejected candidate has a value larger than the current value, so it fails Armijo for every $c_1 > 0$. At each accepted candidate the decrease exceeds one third of the predicted decrease, while the new directional derivative is positive and the old one is negative. Hence Armijo holds for $c_1 \le 1/3$ and weak Wolfe holds for every $c_2 > 0$. Thus the displayed lengths are outputs of the specified search for the entire parameter range in Theorem 10; no accepted-length sequence is assumed. The routine never needs a Wolfe rejection on this example.

Figure 2 displays this unit-first geometry: phase zero accepts its first query, while the next two phases bisect past the rejected candidates before reaching their accepted points. Panel (b) includes all nine queries; panel (c) separates the accepted points near the wrong corner. Line styles distinguish the three phases in grayscale, and the accepted lengths are labeled explicitly.

The finite calculation becomes an infinite run through homogeneity. A smaller copy has the same active slopes, and its inverse matrix produces proportionally smaller steps. Both sides of every line-search comparison have

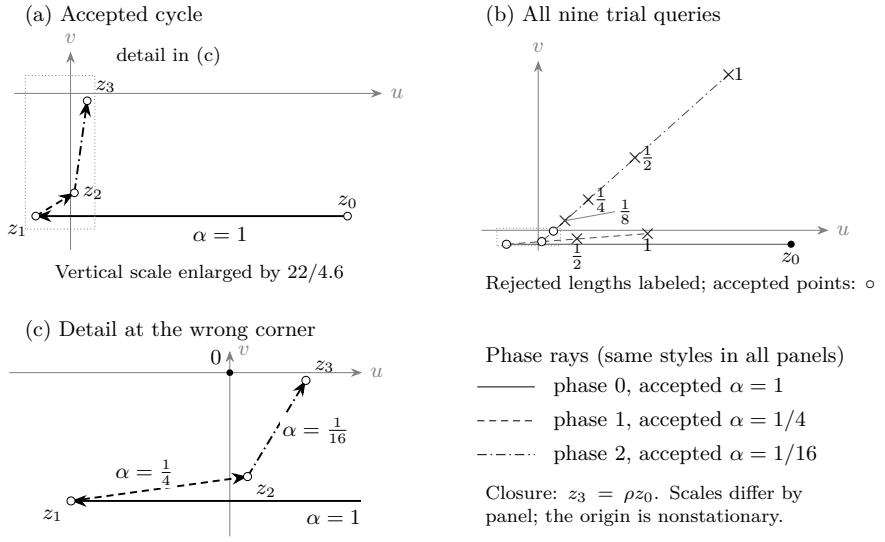


Fig. 2 The accepted three-step cycle, every queried candidate, and an enlargement at the nonstationary origin. Solid, dashed and dash–dot rays mark phases $0, 1, 2$, whose accepted lengths are $1, 1/4, 1/16$. The panels use different horizontal and vertical scales; the vertical-to-horizontal ratios are 22/4.6, 2.5/4.2 and 23/21 in (a), (b) and (c). Coordinates are rendered from the rational center of the certified root box; exact cycle closure and trial decisions follow from Appendix C. This drawing is not evidence of stable-manifold membership for the perturbed construction.

the same scale. For every fixed pair $0 < c_1 \le 1/3$, $c_1 < c_2 < 1$, the two cycle closures therefore give the exact induction

$$x_{3m+j} = \rho^m z_j, \qquad H_{3m+j} = \rho^m M_j, \qquad j = 0, 1, 2. \tag{26}$$

Each query in phase j is ρ^m times its base query. Gradients are unchanged, function differences and gradient–step products scale by ρ^m, and (19) scales the updated matrix by ρ^m. Thus every branch choice and every rejection is preserved, as well as acceptance. The iterates converge to zero, but the direction $w = (-1, 16)^T$ satisfies

$$g_0^T w = g_1^T w = -1, \qquad g_2^T w = -c_\circ + 16 d_\circ < -1.$$

Hence $G(tw) = -t$ for $t \ge 0$, so this limit is nonstationary.

Corollary 8 (Coercive polyhedral global failure) *For every fixed pair $0 < c_1 \le 1/3$, $c_1 < c_2 < 1$, the same deterministic run, with $H_0 = I$, fails for*

$$\overline{G}(u, v) = \max\{u, -u - v/8, c_\circ u + d_\circ v, v - 17\}.$$

This function is coercive and has unique minimizer $(-1, 16)$, with value -1, while the run converges to the nonstationary origin with value zero.

Proof At every query, (25) and $0 < \rho < 1$ give $v - 17 < 0 < G(u, v)$, so the fourth branch is inactive. The convex combination of branches $0, 1, 3$ with weights $8/17, 8/17, 1/17$ is identically -1. Equality in this lower bound forces all three branches to equal -1, hence $(u, v) = (-1, 16)$; the remaining branch is strictly smaller there. If $\overline{G}(u, v) \le t$, then $u \le t$, $v \le t + 17$ and $-u - v/8 \le t$, which also give lower bounds on both coordinates. Every sublevel is therefore compact. □

This already excludes global convergence from every start for coercive finite polyhedral objectives under this routine. Its wrong corner is away from the optimizer, so it is consistent with the local finite-partition convergence theorem. To rule out a local neighborhood guarantee, we need bad corners arbitrarily near the optimizer of the *same* function.

6.3 Matching the spatial scale, objective scale and initial matrix

A change of scale also changes which inverse matrix represents the same search direction. This simple covariance explains both local constructions. Suppose, near the current and queried points under consideration, that

$$F(\bar{x} + \lambda z) = C + \sigma \mathcal{G}(z), \qquad \lambda, \sigma > 0. \tag{27}$$

Differentiation gives $g_F = (\sigma/\lambda) g_{\mathcal{G}}$. Consequently, for the same candidate length, the corresponding inverse matrix and step are

$$H_F = \frac{\lambda^2}{\sigma} H_{\mathcal{G}}, \qquad p_F = \lambda p_{\mathcal{G}}, \qquad s_F = \lambda s_{\mathcal{G}}. \tag{28}$$

At an accepted pair $y_F = (\sigma/\lambda) y_{\mathcal{G}}$. Thus $y_F^T s_F = \sigma y_{\mathcal{G}}^T s_{\mathcal{G}}$; the two projection factors in the inverse update are identical, and its rank-one term is multiplied by λ^2/σ. This proves the matrix relation in (28) again after the update. Function differences and all gradient–step products are multiplied by $\sigma > 0$, so the Armijo and weak-Wolfe comparisons, including their strict signs, agree.

This is a covariance calculation, not an existence theorem for a local trajectory. To transfer a particular run, the local model must cover the current point and every candidate the search actually visits. Under that containment, induction on the bracket decisions transfers the full trial tree, and induction on accepted steps transfers the matrices. We verify that containment separately in each construction below.

The choice $\sigma = \lambda^2$ is decisive: it leaves the initial matrix unchanged. In the first embedding, the corner's objective height is a_ℓ^3 and its affine slopes are $a_\ell^2 g_i$. Taking spatial scale $\lambda = a_\ell^2$ gives $\sigma = a_\ell^4 = \lambda^2$, so every copy starts with I. Its affine neighborhood has width proportional to a_ℓ, much larger than the entire copied trial tree, whose width is proportional to a_ℓ^2. This separation of scales makes all-query containment possible.

6.4 Convex interpolation that leaves room for every query

We now construct the scalar functions that implement this design. Their node values are a_ℓ^3 and their node slopes are a_ℓ^2. A monotone derivative connects neighboring nodes and is constant on a neighborhood of each node. The constant regions create the local affine corner; the cubic transition makes the second derivative continuous and globally Lipschitz even as the nodes accumulate at zero. The following formulas fix the construction completely.

Let $\theta = 1/16$, $a_\ell = \theta^\ell$, $\varepsilon = 1/128$, and $b_* = 2c_\circ - 32d_\circ \in (33, 34)$. For $b \in \{2, b_*\}$ define

$$t_b = \frac{1 - (1 + \theta + \theta^2)/b}{1 - \theta^2}, \qquad v = \frac{\xi - t_b + \varepsilon}{2\varepsilon}.$$

$$P_b(\xi) = \begin{cases} 0 & \xi \le t_b - \varepsilon, \\ 3v^2 - 2v^3 & |\xi - t_b| < \varepsilon, \\ 1 & \xi \ge t_b + \varepsilon, \end{cases} \qquad J_b(\xi) = \int_0^\xi P_b(t)\, dt.$$

The middle formula for J_b is $2\varepsilon(v^3 - v^4/2)$; its outer formulas are zero and $\xi - t_b$. For $b\theta a_\ell \le x \le b a_\ell$, set

$$\varphi_b(x) = (\theta a_\ell)^3 + b a_\ell^3 (1 - \theta)\{\theta^2 \xi + (1 - \theta^2) J_b(\xi)\}, \qquad \xi = \frac{x/(b a_\ell) - \theta}{1 - \theta}. \tag{29}$$

Complete the definition by

$$\varphi_b(x) = \begin{cases} 1 - 3(x+1) + 3(x+1)^2 & x < -1, \\ -x^3 & -1 \le x \le 0, \\ 1 + x - b & x \ge b. \end{cases}$$

Lemma 5 (Scalar interpolation) *The functions φ_b are convex, coercive and globally $C^{2,1}$, are nonnegative, and vanish only at zero. They satisfy*

$$0 \le \varphi_b'' \le \max\{6, 102/b\}, \tag{30}$$

$$\varphi_b(b a_\ell + h) = a_\ell^3 + a_\ell^2 h \qquad (|h| \le a_\ell/2). \tag{31}$$

Their second derivatives are globally Lipschitz with constant

$$L_b = \max\{6, 139264/(5b^2)\}.$$

Proof The ramp is monotone, C^1, and satisfies $0 \le P_b' \le 96$ and $\int_0^1 P_b = 1 - t_b$. On a positive interval,

$$\varphi_b' = a_\ell^2\{\theta^2 + (1 - \theta^2) P_b(\xi)\}, \qquad \varphi_b'' = \frac{a_\ell(1 + \theta)}{b} P_b'(\xi).$$

The identity $\theta^2 + (1 - \theta^2)(1 - t_b) = (1 + \theta + \theta^2)/b$ gives matching values $\varphi_b(b a_\ell) = a_\ell^3$. The derivative is constant near both ends of each interval, so the first two derivatives also match, with values a_ℓ^2 and zero at $b a_\ell$.

As $x \downarrow 0$, the bounds $a_\ell \le x/(b\theta)$ on its interval give $\varphi_b(x) = O(x^3)$, $\varphi_b'(x) = O(x^2)$ and $\varphi_b''(x) = O(x)$. These match the negative cubic at zero. The formulas also match through second order at -1 and b. The derivative is negative on the negative half-line and positive on the positive half-line; the outer formulas diverge to infinity. The second derivative is nonnegative on every piece: a positive multiple of P_b' on the positive intervals, and 6, $-6x$ and 0 for $x < -1$, $-1 \le x \le 0$ and $x \ge b$. This proves convexity, coercivity and the minimum assertions.

The constant-derivative lengths below and above ba_ℓ, divided by a_ℓ, are

$$b(1-\theta)(1-t_b-\varepsilon), \qquad \frac{b(1-\theta)(t_b-\varepsilon)}{\theta}.$$

Both exceed $1/2$ for $2 \le b \le 34$, by substitution in the displayed rational formulas; the affine outer extension covers the upper side when $\ell = 0$. This proves (31). Finally $|P_b''| \le 3/(2\varepsilon^2)$ in each ramp, so differentiation once more gives $|\varphi_b'''| \le 139264/(5b^2)$ there. The negative cubic has third derivative of magnitude six. Continuity of φ_b'' permits summing the Lipschitz estimates over finitely many joins away from zero; taking limits at zero covers the infinitely many accumulating joins and gives the global bound. □

6.5 One objective and a uniform induction over all queries

Define

$$F(u,v) = \max\{\varphi_2(u), \varphi_2(-u-v/8), \varphi_{b_*}(c_\circ u + d_\circ v), v\}. \tag{32}$$

The Hessian of each of the first three branches is a nonnegative scalar multiple of $g_i g_i^T$. Equation (30) gives the upper bounds 51, $51(65/64) < 52$, and 12, respectively, on their operator norms; the last branch has zero Hessian. Their Hessians are globally Lipschitz because φ_b'' is Lipschitz and the arguments are linear. The first two scalar arguments are independent, so their coercivity makes every sublevel of F compact. Nonnegativity and their unique scalar zeros show $F(x) = 0$ exactly at $x = 0$. At zero the first three gradients vanish and the last is $(0,1)^T$, giving $F'(0;h) = \max\{0, h_2\}$; thus the minimizer is nonsmooth.

Let $\bar{x}_\ell = a_\ell(2,-32)^T$. The first two scalar arguments at $\bar{x}_\ell$ equal $2a_\ell$, and the third equals $b_* a_\ell$. Consequently

$$F(\bar{x}_\ell + h) = a_\ell^3 + a_\ell^2 G(h) \quad \text{if } |g_i^T h| \le a_\ell/2 \quad (i = 0,1,2). \tag{33}$$

Indeed, the first three branches are on their affine regions and each has value at least $a_\ell^3/2$. The first two slab inequalities imply $|h_1| \le a_\ell/2$ and $|h_2| \le 8a_\ell$, so the fourth branch is negative. In particular $F(\bar{x}_\ell) = a_\ell^3$ and $F'(\bar{x}_\ell; (-1,16)^T) = -a_\ell^2 < 0$.

Proof (Theorem 10) For $\ell \geq 1$ take $x_0^{(\ell)} = \bar{x}_\ell + a_\ell^2 z_0$ and $H_0 = I$. We claim that

$$x_{3m+j}^{(\ell)} = \bar{x}_\ell + a_\ell^2 \rho^m z_j, \qquad H_{3m+j}^{(\ell)} = \rho^m M_j, \qquad j = 0, 1, 2. \tag{34}$$

At a current point of this form its gradient is $a_\ell^2 g_j$. Every trial on the finite candidate list therefore has the form $\bar{x}_\ell + a_\ell^2 \rho^m z_{\text{trial}}$. Equation (25) gives

$$|g_i^T (a_\ell^2 \rho^m z_{\text{trial}})| < 3a_\ell^2 \leq 3a_\ell/16 < a_\ell/2.$$

Thus (33) is valid at every candidate query, not only the final accepted one. Unique activity is preserved, and

$$F(\bar{x}_\ell + a_\ell^2 \rho^m z_{\text{trial}}) = a_\ell^3 + a_\ell^4 \rho^m G(z_{\text{trial}}), \qquad \nabla F = a_\ell^2 g_{\text{active}}.$$

These identities instantiate (27) with $\lambda = a_\ell^2$, $\sigma = a_\ell^4$ and $C = a_\ell^3$. The containment just proved applies before each candidate is tested, so the scaling argument transfers every bracket decision. Equivalently, the additive constant cancels in Armijo and every function difference and gradient–step product is the seed quantity times $a_\ell^4 \rho^m > 0$. The actual search traverses exactly the candidate list and makes the same strict decisions. In (19), the projection factors are unchanged and the rank-one term has factor ρ^m. The cycle closures prove the next point and matrix identities, completing the induction.

The finite matrices are positive definite. Equation (34) gives convergence to $\bar{x}_\ell$ and strict descent to $a_\ell^3 > 0$. Since the base query set is finite, the norm of every query is bounded by

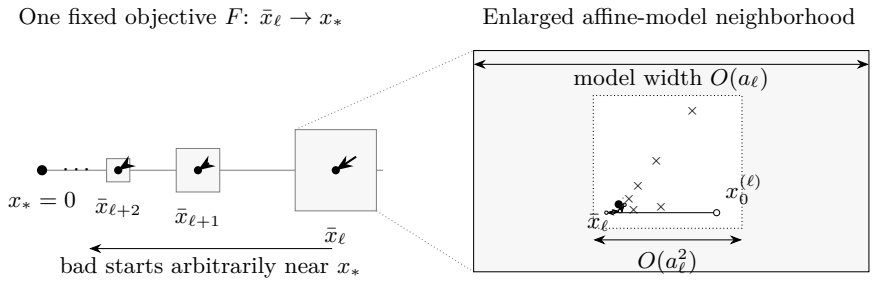
$$a_\ell \|(2, -32)\| + a_\ell^2 \max_{\text{base points and queries}} \|z\|,$$

which tends to zero. The starting points also tend to zero. All other claims were proved directly from (32). □

Figure 3 makes the local quantifier explicit: one function contains every shrinking copy. For each fixed copy the run converges to its own wrong center, while the starting points and all their queries can be made arbitrarily close to the same true minimizer.

6.6 What changes when positive lower curvature is required

The flat affine regions in the first construction explain why its convex branches have no uniform positive lower curvature. Adding the same positive quadratic to each branch restores that curvature, but changes the dynamics. The old exact cycle can no longer be copied without adjustment. The next result constructs a nearby trajectory by a stable-manifold argument.



Schematic: locations and relative scales exaggerated. This panel concerns the $C^{2,1}$ embedding only.

All accepted and rejected queries fit inside. Their contraction copies accumulate at $\bar{x}_\ell$, not at x_*.

Fig. 3 The $C^{2,1}$ local embedding, schematically. In one fixed objective, bad centers $\bar{x}_\ell \to x_*$ have affine-model neighborhoods of width $O(a_\ell)$. Each complete copied query orbit fits inside a region of size $O(a_\ell^2)$ and converges to $\bar{x}_\ell$. The inset shows the seed and two contraction copies as a finite illustration of the infinite induction. Locations, neighborhood shapes and relative scales are schematic, not a plot of the infinitely defined objective. The lower-curvature construction uses the different spatial factor a_ℓ, as explained below.

Theorem 11 (Independent local failure with positive lower curvature) *There is a finite strongly convex function $F : \mathbb{R}^2 \to \mathbb{R}$ with compact sublevels and a unique nonsmooth minimizer at zero, with fixed starting points $x_0^{(\ell)} \to 0$, all with $H_0 = I$, such that for every fixed pair $0 < c_1 \le 1/3$, $c_1 < c_2 < 1$ the deterministic unit-first bracketing implementation of Section 6 produces infinite full-BFGS runs, the same for every such pair. They converge to nonstationary points $\bar{x}_\ell \neq 0$, with $\bar{x}_\ell \to 0$ and $F(x_k^{(\ell)}) \downarrow F(\bar{x}_\ell) > F(0)$. Every accepted and rejected query is differentiable, the accepted lengths repeat $1, 1/4, 1/16$, and the supremum of all query norms in the ℓth run tends to zero.*

The objective can be a maximum of four globally $C^{1,1}$, uniformly strongly convex branches with globally Lipschitz gradients. There are fixed $0 < \underline{\mu} \le \overline{L} < \infty$ such that

$$\underline{\mu} I \preceq \nabla^2 F(x) \preceq \overline{L} I \quad \text{at every twice-differentiable point.}$$

The first three branches in this construction are not twice differentiable at zero.

In Theorem 11 and Table 3, a Hessian is the classical one: the derivative of the gradient map, with that map defined in a neighborhood of the point. A pointwise second-order expansion alone is not the convention.

The proof in Appendix D has the same sequence of tasks as the first construction. It starts from the explicit strongly convex objective $Q_\mu(z) = \mu\|z\|^2/2 + G(z)$, whose true minimizer is away from the wrong corner. A normalized three-step return map has the original seed as a hyperbolic boundary fixed point. Its stable manifold contains positive scale states, yielding an actual perturbed bad orbit. The seed's strict finite margins ensure that the same accepted and rejected trial lists persist along that orbit, for every fixed pair

Branch regularity	Uniform curvature	At the minimizer
Four convex $C^{2,1}$ branches	$0 \preceq \nabla^2 f_i \preceq 64I$ globally	All branch Hessians exist and are Lipschitz; no uniform positive lower bound
Four uniformly strongly convex $C^{1,1}$ branches	Positive lower bound and globally Lipschitz branch gradients	The first three branches are not twice differentiable

Table 3 The two independent local failure regimes. Each uses one fixed objective and $H_0 = I$, with bad starts arbitrarily close to its unique minimizer; all actual accepted and rejected queries are differentiable and remain in neighborhoods shrinking to that minimizer. Both hold for every fixed $0 < c_1 \leq 1/3$, $c_1 < c_2 < 1$ (Theorems 10 and 11).

$0 < c_1 \leq 1/3$, $c_1 < c_2 < 1$. No rounded initial state is asserted to lie on the stable manifold.

For the local embedding of Appendix D, with its own $a_\ell = 128^{-\ell}$, centers $\bar{x}_\ell = a_\ell(2, -32)^T$ and constant $C_* = 1 + \mu\|(2, -32)\|^2/2$, the scalar node values are now a_ℓ^2 and the node slopes a_ℓ. The resulting model is $F(\bar{x}_\ell + a_\ell z) = a_\ell^2(C_* + Q_\mu(z))$, so $\lambda = a_\ell$, $\sigma = a_\ell^2$ and again $\lambda^2/\sigma = 1$. We choose one entire perturbed orbit small enough for every query to fit the affine scalar regions. Its one initial SPD matrix is then used in every copy; one fixed affine change makes it I. The faster variation of the scalar derivative retains global $C^{1,1}$ regularity but prevents second differentiability at the minimizer, where the copies accumulate. This is why the theorem does not supply a finite strongly convex C^2 local counterexample.

The two constructions (Table 3) separate upper branch regularity from positive lower curvature. Their common mechanism is a local copy of an actual bad query tree; the different scalar scales determine which regularity survives at the minimizer. The remaining convergence questions are discussed next.

7 Synthesis, verification and the remaining questions

The mixed benchmark $u^2 + |v|$ has a whole-sequence convergence answer within an arbitrary-dimensional centered-family theorem. Finite polyhedral geometry supplies a local inward model, while a moving field handles the three-quadratic example's curved switches. In each case comparison energies use the principal-minor hierarchy to detect a remaining objective gap.

Why the counterexamples evade these hypotheses. The distinctions can be read from the constructed objectives. The homogeneous seed satisfies $G(t(-1, 16)) = -t$ for $t \geq 0$, so it is unbounded below; its coercive cap contains the affine term $v - 17$ and is not homogeneous. Neither belongs to the coercive homogeneous class of Theorem 6. In the local embeddings, wrong vertices recur at infinitely many shrinking scales around the same minimizer. They cannot be removed by taking a smaller neighborhood, and the resulting switching geometry is not

resolved by a finite local polyhedral-cell partition as in Theorem 7. Finitely many smooth formulas do not by themselves supply that partition.

There is also a direct test of the planar hypotheses. In (32), all four branches are active at zero, with indexed gradients $0, 0, 0, e_2$, where $e_2 = (0, 1)^T$. Equation (54) gives the same pattern before the fixed change of coordinates of Appendix D: the common quadratic has zero gradient at zero, and $\psi_b'(0) = 0$. After that change, $x = Az$ with $A = (H_0^{\text{seed}})^{1/2}$, the gradients are $0, 0, 0, A^T e_2$, with $A^T e_2 \neq 0$. Thus the repeated zero gradients violate affine independence, and a combination with positive weight λ_3 on the fourth branch equals $\lambda_3 e_2 \neq 0$ (or $\lambda_3 A^T e_2 \neq 0$ after that change). The strictly positive optimal weights in Corollary 7 therefore fail as well. Three of the four lower-curvature branches also lose second differentiability at zero. Positive strong convexity alone supplies neither the missing regularity nor this nondegeneracy.

The secant calculation explains how curvature and the failure mechanism coexist. For $Q_\mu(x) = \mu\|x\|^2/2 + G(x)$, suppose the differentiable endpoints x and $x + s$ have active slopes g_i and g_j. Then exactly

$$y = \big(\mu(x + s) + g_j\big) - \big(\mu x + g_i\big) = \mu s + (g_j - g_i). \tag{35}$$

Along the perturbed orbit of Proposition 1, $s \to 0$ while the three distinct slopes continue to cycle. The positive-curvature contribution therefore vanishes with the step, while the branch jumps persist. The stable-manifold and strict-query arguments establish that this decomposition is realized by the specified algorithm; a branch change alone does not imply failure.

These comparisons explain sufficient hypotheses separating the proved positive classes from the examples. They do not make affine independence, positive multipliers or a comparison certificate necessary for convergence, and they do not turn every nonoptimal affine-critical point or accumulating singularity into a counterexample. The local question for finitely many uniformly strongly convex C^2 branches remains open here outside the nondegenerate planar case of Corollary 7. The examples concern the specified weak-Wolfe implementation with $0 < c_1 \leq 1/3$ and exceptional deterministic starts; they do not settle Lewis–Overton's almost-sure challenge under its additional hypotheses [1, Challenge 7.1].

Formal verification and reproducibility. The complete companion supplies pinned Lean sources, exact theorem contracts and their correspondence to the manuscripts, compiler, axiom and kernel-replay audits, and versioned source/build records. The proofs and rational certificates needed for the article's main results are also contained here. Formal checking and author-side correspondence review have distinct roles; neither certifies publication priority or floating-point robustness.

AI assistance. Large language models assisted theorem development, proof reconstruction, formalization, literature searches, software and writing. The companion documents this substantive research assistance and the automated author-side reviews.

A The BFGS compression calculation

Lemma 6 *Under the assumptions of Theorem 1, let Z have j orthonormal columns, $1 \le j \le n$. Then*

$$\det(Z^T H_+ Z) \le \frac{\det \mathrm{Gram}_H(y, Z)}{\beta^2 q} + \frac{\alpha}{\beta} \det(Z^T H Z).$$

For $j = n$, $\det H_+ = (\alpha/\beta) \det H$.

Proof Put $v = Hg$, $\kappa = g^T H y = -\beta q$ and $P = I - vy^T/\kappa$. Since $y^T s = \alpha\beta q$, the update becomes

$$H_+ = \mathcal{F} + \frac{\alpha}{\beta q} v v^T, \qquad \mathcal{F} = PHP^T.$$

Write

$$A = Z^T H Z \succ 0, \quad a = Z^T H g, \quad b = Z^T H y, \quad r = y^T H y,$$
$$u = a^T A^{-1} a, \quad w = a^T A^{-1} b, \quad z = b^T A^{-1} b.$$

Then

$$Z^T \mathcal{F} Z = A - [a\ b] \begin{pmatrix} -r/\kappa^2 & 1/\kappa \\ 1/\kappa & 0 \end{pmatrix} [a\ b]^T.$$

The identity $\det(I + UV) = \det(I + VU)$ reduces its determinant to a 2×2 determinant. Expanding gives

$$\det(Z^T \mathcal{F} Z) = \frac{\det A}{\kappa^2} \big((\kappa - w)^2 + u(r - z) \big). \tag{36}$$

With $\mathrm{Gram}_H(g, Z) = [g\ Z]^T H [g\ Z]$ and $\mathrm{Gram}_H(g, y, Z) = [g\ y\ Z]^T H [g\ y\ Z]$, Schur complements give, independently,

$$\det \mathrm{Gram}_H(y, Z) = (r - z) \det A,$$
$$\det \mathrm{Gram}_H(g, y, Z) = \big((q - u)(r - z) - (\kappa - w)^2 \big) \det A.$$

Both are Gram determinants and hence nonnegative, including when their column families are dependent. Combining these identities yields

$$\det(Z^T \mathcal{F} Z) = \frac{q \det \mathrm{Gram}_H(y, Z) - \det \mathrm{Gram}_H(g, y, Z)}{\beta^2 q^2} \le \frac{\det \mathrm{Gram}_H(y, Z)}{\beta^2 q}.$$

Adding $\rho a a^T$ to the compression changes the coefficient r/κ^2 by ρ. The same 2×2 expansion therefore changes its determinant by exactly $\rho \det A\, u$. No invertibility of $Z^T \mathcal{F} Z$ is used. Since $Z^T v = a$, the value $\rho = \alpha/(\beta q)$ gives the compression $Z^T H_+ Z$. The Schur complement of A in $\mathrm{Gram}_H(g, Z)$ gives $u \le q$ (and $u \ge 0$ since $A \succ 0$), so the increase is at most $(\alpha/\beta) \det A$. This proves the inequality.

When $j = n$, Z is an orthogonal matrix, $A^{-1} = Z^T H^{-1} Z$, and $u = q$, $w = \kappa$, $z = r$. The right side of (36) then vanishes, while the rank-one increase is exactly $(\alpha/\beta) \det A$. Orthogonal invariance gives the stated determinant identity. □

B Constructing the planar comparison field

This appendix supplies the geometric ingredient used in Theorem 8. The construction applies near a compact regular part of one positive level. It does not assume that the BFGS iterates converge to one point of that level.

Lemma 7 (One field with all endpoint estimates) *Let $F : \mathbb{R}^2 \to \mathbb{R}$ be finite and convex with minimum F_*, and let $C \subset \{F = \ell\}$ be a nonempty compact set, where $\ell > F_*$. Suppose the boundary of $\{F \le \ell\}$ contains no nontrivial segment and every point of C is non-affine-critical and has the smooth or regular convex two-branch structure in Theorem 8. Fix valid Armijo–weak-Wolfe constants. There exist an open neighborhood $U \supset C$, a compact buffer $Q \supset \overline{U}$, a field w defined on a neighborhood of Q, and constants $\eta, \kappa, \delta > 0$, $C_0, C_1, L \ge 0$ such that:*

1. *The field is locally Lipschitz, bounded on Q, and every $g \in \partial F(p)$, $p \in Q$, satisfies $g^T w(p) \ge \eta$. All subgradients at a fixed point have the same pairing with $w(p)$.*
2. *For every eligible pair $x, z \in U$ with $\|s\| < \delta$, its entire segment lies in Q and*

$$|\det(s, w(x + \theta s))| \ge \kappa \|s\| \quad (0 \le \theta \le 1), \qquad (y^T w(z))^2 / t \le C_0.$$

3. *For every such pair there are $\lambda > 0$ and $\beta \in \mathbb{R}$ with*

$$w(z) = \lambda w(x) + \beta s, \qquad |\log \lambda| \le C_1 d, \qquad d \le 1 \implies \lambda^2 \le 1 + Ld.$$

All neighborhoods, constants and the field are chosen before the endpoint pair or any BFGS matrix.

Proof Corner motion. At a nonsmooth point of C, the local branch-value map $x \mapsto (f_1(x), f_2(x))$ has nonsingular derivative. The inverse function theorem gives a C^2 curve $p_j(t)$ characterized by $f_1(p_j(t)) = f_2(p_j(t)) = t$ for t near ℓ. Its C^1 velocity $v_j(t) = p_j'(t)$ satisfies

$$\nabla f_1(p_j(t))^T v_j(t) = \nabla f_2(p_j(t))^T v_j(t) = 1. \tag{37}$$

In particular, this chart has only one nonsmooth point at level ℓ. A smooth point has a neighborhood with no nonsmooth points. A finite cover of C by smaller buffered charts therefore contains only finitely many level-ℓ corners. After shrinking their charts, the neighborhood of C, and the interval of levels, every nearby nonsmooth point lies on one of these finitely many curves. This is a compactness consequence of the charts, rather than an assumed finite bound on future branch changes.

Joining their directions. At a corner, the two active gradients generate its cone of supporting normals. Its unit directions form a compact arc of width less than π; by (37), every such unit normal n has $n^T v_j(t) > 0$. Normal arcs at distinct level-ℓ corners are disjoint: a shared supporting normal would expose

both points and their full segment in the boundary of $\{F \le \ell\}$. The strict-contour hypothesis excludes this. The finitely many arcs have positive angular separation.

Choose fixed smooth angular cutoffs $0 \le \chi_j \le 1$ with disjoint supports, equal to one on slightly enlarged normal arcs. Shrink the level interval so that these plateaus contain the whole moving arcs and each support still satisfies $n^T v_j(t) > 0$ uniformly. With $n(\theta) = (\cos\theta, \sin\theta)$ and $T(\theta) = (-\sin\theta, \cos\theta)$, define

$$h(t,\theta) = 1 + \sum_j \chi_j(\theta)\big(n(\theta)^T v_j(t) - 1\big). \tag{38}$$

Because at most one cutoff is nonzero, this is a convex combination of positive quantities. Thus h is bounded and bounded away from zero on the chosen compact level interval. It is C^1 in t, smooth in θ, and its derivatives used below are bounded. No convex-support-function property of h is needed.

At a smooth point, put $n(\theta(x)) = \nabla F(x)/\|\nabla F(x)\|$ and define

$$w(x) = h(F(x),\theta(x))n(\theta(x)) + h_\theta(F(x),\theta(x))T(\theta(x)). \tag{39}$$

All expressions are periodic, so no global choice of angle is required. Near the jth switching curve the relevant cutoff equals one and all others vanish. Consequently $h = n^T v_j$, $h_\theta = T^T v_j$, and

$$w(x) = v_j(F(x)). \tag{40}$$

This holds on both smooth sides in a whole neighborhood of the curve. Use the same formula on the curve itself. It provides a continuous, locally Lipschitz extension across the gradient jump, while the field is C^1 elsewhere. Smaller compact buffers give uniform field and derivative bounds. At a smooth point,

$$\nabla F(x)^T w(x) = \|\nabla F(x)\| h(F(x),\theta(x)) > 0;$$

at a corner every subgradient pairs to one by (37). On sufficiently small buffers of C, these pairings have a uniform positive lower bound. One way to see the needed uniform gradient bound away from zero is the supporting inequality $\|g\|\|x - x_*\| \ge F(x) - F_*$ on the compact buffer, where x_* is any minimizer and the value gap stays positive. This proves the visibility and common-pairing assertions.

The specific angular formula has a crucial derivative cancellation:

$$\partial_\theta(hn + h_\theta T) = (h + h_{\theta\theta})T. \tag{41}$$

Thus at a smooth point the derivative of w maps the tangent line of a level curve into that same line. In the orthonormal basis (T, n),

$$Dw = \begin{pmatrix} a & b \\ 0 & e \end{pmatrix}$$

with bounded coefficients. The formula $hn + h_\theta T$ is classical in normal-angle geometry [17]; Bogosel gives its explicit form (2022 preprint, arXiv:2203.06981v1,

Section 2, equation (4)). Here its derivative cancellation is used for a comparison field, without a convexity requirement on h.

Transversality of eligible steps. Choose nested neighborhoods so that $\overline{U}$ lies in the interior of Q, with all preceding bounds valid on a larger buffer. Sufficiently close pairs in U have their segments in Q. If the claimed determinant lower bound failed at arbitrarily short lengths, compactness would give coalescing endpoints tending to $p \in Q$, normalized steps $s/\|s\| \to \hat{s}$, and segment points at which the normalized determinant tends to zero. Bounded subgradients and their closed graph give endpoint gradient limits $\bar{g}, \bar{h} \in \partial F(p)$. Continuity of w makes $\hat{s}$ parallel to $w(p)$. Positivity of the common pairing and $d > 0$ force $\hat{s} = -w(p)/\|w(p)\|$. Hence

$$d/\|s\| \longrightarrow -\bar{g}^T \hat{s} > 0, \qquad t/\|s\| \longrightarrow (\bar{h} - \bar{g})^T \hat{s} = 0,$$

contradicting $t \geq (1 - c_2)d$. This proves uniform transversality, including every point of the segment, after reducing δ.

Transport charged to decrease. Write $q(\theta) = x + \theta s$ ($\theta \in [0, 1]$ is now a segment parameter, not a normal angle), $\varphi(\theta) = F(q(\theta))$ and $D(\theta) = \det(s, w(q(\theta)))$. Transversality makes D nonzero with constant sign on $[0, 1]$. At a smooth point the basis (T, n) has determinant -1. Writing $s = s_T T + s_n n$, direct calculation yields

$$\det(s, Dw\, s) = s_n \big[(a - e)s_T + b s_n\big]. \tag{42}$$

Since $D'(\theta) = \det(s, Dw\, s)$ there, the numerator of D'/D has magnitude at most a constant times $|s_n|\|s\|$. Since $|D| \geq \kappa\|s\|$ and $\varphi' = \|\nabla F\|\, s_n$, the quotient is bounded by a constant times $|\varphi'|$. At a corner, (40) gives $w \circ q = v_j \circ \varphi$. The convex function φ has a right derivative at every $\theta < 1$, and the one-sided chain rule gives the same bound there. Therefore, for $H(\theta) = \log|D(\theta)|$,

$$|H'_+(\theta)| \leq A|\varphi'_+(\theta)| \leq A\big(\varphi'_+(\theta) + 2d\big), \qquad 0 \leq \theta < 1.$$

The last inequality uses monotonicity of the convex right derivative and $\varphi'(0) = -d$.

For clarity, the one-sided comparison needed here requires no finite count of switching intersections. If a continuous function u on $[0, 1]$ has $u'_+ \leq 0$ on $[0, 1)$, then $u(1) \leq u(0)$. Otherwise choose $\epsilon > 0$ so that $u(1) - \epsilon > u(0)$. The continuous function $u(\theta) - \epsilon\theta$ has a minimum at some point before 1, where its right derivative must be nonnegative, contradicting the strictly negative bound. Apply this fact to both

$$\pm\big(H(\theta) - H(0)\big) - A\big(\varphi(\theta) - \varphi(0) + 2d\theta\big).$$

Both start at zero and have nonpositive right derivative. Since $\varphi(1) \leq \varphi(0)$,

$$|H(1) - H(0)| \leq 2Ad.$$

Set $\lambda = D(1)/D(0) > 0$. Because $s, w(x)$ are linearly independent, solving in that basis gives $w(z) = \lambda w(x) + \beta s$; the preceding bound is $|\log\lambda| \leq 2Ad$.

For $0 < d \le 1$, convexity of the exponential on $[0,1]$ gives $\lambda^2 \le \exp(4Ad) \le 1 + (\exp(4A) - 1)d$. This establishes the transport constants.

The secant correction. A finite buffered chart cover supplies a common Lipschitz constant for the branch gradients. First suppose the endpoints use the same smooth branch f. On a larger convex chart, with Lipschitz constant $M > 0$, sufficiently close buffered endpoints satisfy

$$\|\nabla f(z) - \nabla f(x)\|^2 \le 2M\big[f(z) - f(x) - \nabla f(x)^T(z - x)\big].$$

Indeed, subtract the tangent plane at x, then use the descent estimate at z for the step $-(\nabla f(z) - \nabla f(x))/M$. This step stays in the larger chart for close endpoints, and convexity makes the resulting function value no smaller than its value at x. This proves the displayed inequality. It is the local convex-gradient estimate underlying cocoercivity [2]; compare (D. Wachsmuth and G. Wachsmuth, 2022 preprint, arXiv:2204.00282v1, Theorem 2.5). Since the branch values and gradients at both endpoints agree with those of F, the bracket is at most t. Boundedness of w then bounds $(y^T w(z))^2/t$ uniformly.

For different endpoint branches i, j in a two-branch chart, the segment contains a tie point p. There $(\nabla f_j(p) - \nabla f_i(p))^T w(p) = 0$. The branch gradients and w are Lipschitz, so

$$|y^T w(z)| \le A_1 \|s\|.$$

Put $r = \nabla f_j(z) - \nabla f_j(x)$, so $\|r\| \le M\|s\|$. Then

$$\nabla f_i(x)^T s = -d, \qquad \nabla f_j(x)^T s = t - d - r^T s.$$

The two gradient rows are uniformly independent on the smaller chart. Their inverse bound and eligibility imply

$$\|s\| \le A_2(2d + t + M\|s\|^2) \le A_2 \left(1 + \frac{2}{1 - c_2}\right) t + A_2 M \|s\|^2.$$

For sufficiently small $\|s\|$, absorb the last term to obtain $t \ge a_0\|s\|$ with $a_0 > 0$. Consequently $(y^T w(z))^2/t \le A_1^2\|s\|/a_0$, uniformly bounded. A finite buffered cover ensures every sufficiently close pair lies in one of the preceding charts. Taking the common smaller distance threshold finishes all three assertions for the same field. □

C The algebraic seed and its finite rational certificate

This appendix gives the objects and finite inequalities needed for Lemma 4. There are two logically separate tasks. The algebraic equations close the three-step matrix and displacement cycle; the strict inequalities force the specified search to choose that cycle. Solving the equations alone would leave the algorithmic claim unproved. All decimals defining a box are terminating rationals. No rounded point is substituted for a root or an infinite orbit.

C.1 Existence and uniqueness of the defining root

The two unknown coordinates of the third slope are used to enforce the two prescribed dyadic lengths. The matrix calculation in the next subsection first expresses the lengths needed for closure as rational functions $\beta(c_\circ, d_\circ)$ and $\gamma(c_\circ, d_\circ)$. Clearing their positive denominators in $\beta = 1/4$ and $\gamma = 1/16$ gives the following two equations. A contraction certificate then selects one exact solution near a rational center.

Put

$$\begin{aligned} U &= 127c_\circ + 16d_\circ, & V &= 16c_\circ - 127d_\circ - 16, & R &= c_\circ - 8d_\circ, \\ K &= 8c_\circ^2 + 8d_\circ^2 + d_\circ - 8, & T &= c_\circ^2 - c_\circ + d_\circ^2, & Y &= (c_\circ - 1)^2 + d_\circ^2, \end{aligned}$$

and let

$$P(c_\circ, d_\circ) = \begin{pmatrix} 264192RK - UV(U + 129) \\ 16VTK - RUY \end{pmatrix}. \tag{43}$$

The root is specified by $P(c_\circ, d_\circ) = 0$ and $\|(c_\circ, d_\circ) - \bar{r}\|_\infty \le 10^{-30}$, where

$$\begin{aligned} \bar{r}_1 &= 0.318481427627118630646451312502907037179122050236 05, \\ \bar{r}_2 &= -1.025232119675748725481350870013004142845119093184 44. \end{aligned}$$

Lemma 8 (Rational root certificate) *This closed box contains exactly one root. On the box all of $U, V, R, K, T, Y, U + 129$ are positive, and* (20) *holds.*

Proof Let $D = DP(\bar{r})$ and define its inverse by the ordinary two-by-two adjugate formula $A = \operatorname{adj}(D)/\det D$. The following finite rational certificate uses only the coefficients of (43) and the two displayed rationals:

$$\begin{gathered} \det D \neq 0, \qquad \max_i |(AP(\bar{r}))_i| < 10^{-50}, \\ \max_i \sum_{2 \le |\nu| \le 5} |a_{i,\nu}|\, |\nu| \le 7000, \end{gathered} \tag{44}$$

where, for $\nu = (\nu_1, \nu_2)$,

$$a_{i,\nu} = -\frac{1}{\nu_1!\nu_2!} \sum_{j=1}^{2} A_{ij}\, \partial^\nu P_j(\bar{r}).$$

These formulas specify every rational in the certificate: the index set contains the eighteen monomials of degrees two through five. The bounds in (44) follow by substituting the rational center, differentiating the displayed polynomials, and clearing positive denominators in the three comparisons. Thus they are finite integer inequalities, rather than tolerances for a numerically computed root.

For completeness, their analytic consequence follows directly. The centered residual map $N(h) = h - AP(\bar{r} + h)$ has the exact expansion

$$N_i(h) = -(AP(\bar{r}))_i + \sum_{2 \le |\nu| \le 5} a_{i,\nu} h^\nu;$$

its linear part vanishes because $AD = I$. On $\|h\|_\infty, \|k\|_\infty \le r = 10^{-30}$, telescoping the factors of each monomial gives $|h^\nu - k^\nu| \le |\nu| r \|h - k\|_\infty$, since $r \le 1$ and $|\nu| \ge 2$. Hence N has Lipschitz constant at most $7000r$ and

$$\|N(h)\|_\infty \le 10^{-50} + 7000r^2 < r.$$

The closed box is complete, so contraction gives a unique fixed point. Since A is invertible, this is exactly a root of P. This is the standard preconditioned-residual principle; compare Rump [16, Theorem 13.3].

The root box lies in $318/1000 < c_\circ < 319/1000$ and $-1026/1000 < d_\circ < -1025/1000$. Substitution gives positivity of the stated factors. For example,

$$\frac{2397}{100} \le U \le \frac{24113}{1000}, \quad \frac{119263}{1000} \le V \le \frac{59703}{500}, \quad \frac{4259}{500} \le R \le \frac{8527}{1000}.$$

Multiplying positive endpoints in $129R/[V(U + 129)]$ gives $60/1000 < \rho < 61/1000$. In particular all divisions used below are valid. □

C.2 The matrix identity

Proof (the algebraic part of Lemma 4) The first update of the cycle, $\mathcal{B}(I, g_0, g_1, 1) = M_1$, is immediate from $s_0 = (-1, 0)^T$ and $g_1 - g_0 = (-2, -1/8)^T$. For an arbitrary point $(c_\circ, d_\circ)$, not only the root, with the positive factors above, put

$$\beta = \frac{66048RK}{UV(U + 129)}, \quad \gamma = \frac{VTK}{RUY}, \quad \widetilde{\rho} = \frac{2064TK}{U(U + 129)Y}.$$

The rational identities obtained from (19) are

$$\begin{aligned} \mathcal{B}(\mathcal{B}(M_1, g_1, g_2, \beta), g_2, g_0, \gamma) &= \widetilde{\rho} I, \\ -\gamma \mathcal{B}(M_1, g_1, g_2, \beta) g_2 &= \widetilde{\rho}(g_0 - g_2). \end{aligned} \tag{45}$$

Here is a way to check the identities without guessing matrix entries. The middle update is $C + \beta E$, with projected term C and rank-one term βE, because its two projection factors are independent of the step length and its rank-one term is linear in that length. Requiring $(C + \beta E)g_2$ to be parallel to $g_0 - g_2$ gives a linear equation for β; substitution of M_1, g_1, g_2 gives the displayed value. For a unit vector $w \perp (g_0 - g_2)$, substitution gives $w^T(C + \beta E)w = \widetilde{\rho}$. With the displayed γ, the last step is $\widetilde{\rho}(g_0 - g_2)$. Its projection factors are the orthogonal projector $\mathit{\Pi} = ww^T$ onto $(g_0 - g_2)^\perp$, so its projected term is $\mathit{\Pi}(C + \beta E)\mathit{\Pi} = \widetilde{\rho}\,\mathit{\Pi}$, and its rank-one term is $\widetilde{\rho}(I - \mathit{\Pi})$. Their sum is $\widetilde{\rho} I$, proving (45). Each substitution is a rational identity and can equivalently be checked by clearing its nonzero denominators and expanding.

Equations (43) give exactly $\beta = 1/4$ and $\gamma = 1/16$. Moreover,

$$\widetilde{\rho} - \rho = \frac{2064R}{V(U + 129)} \left(\gamma - \frac{1}{16} \right) = 0.$$

This gives the matrix and displacement closures. The steps (22) also satisfy $s_j = -\alpha_j M_j g_j$ with $\alpha_0 = 1$, $\alpha_1 = 1/4$ and $\alpha_2 = 1/16$: for $j = 0, 1$ by substitution, and for $j = 2$ by the second identity in (45). The formula for z_0 gives $s_0 + s_1 + s_2 = -(1-\rho)z_0$. The strict inequalities below imply $(g_{j+1} - g_j)^T s_j > 0$ at each phase. Positive definiteness follows inductively from the full update: its two nonnegative quadratic terms cannot both vanish on a nonzero vector when $y^T s > 0$. □

C.3 Sound enclosure of every query

The remaining task is finite because one cycle has only nine queries. For each query we must certify its unique active branch, the descent direction at the current iterate, and the comparison that accepts or rejects it. A positive derivative at each accepted query supplies weak Wolfe for every permitted curvature constant. Strict value increase at each rejected query makes Armijo fail, so the routine reaches exactly the next displayed bisection. This explains why the certificate below contains more than values at the three accepted points.

An interval $[a, b]$ encloses the corresponding real quantity. Addition, negation, multiplication and reciprocal are evaluated by

$$\begin{aligned} [a,b] + [c,d] &= [a+c, b+d], \qquad -[a,b] = [-b,-a], \\ [a,b][c,d] &= [\min(ac,ad,bc,bd), \max(ac,ad,bc,bd)], \end{aligned}$$

and $[a, b]^{-1} = [1/b, 1/a]$ when $0 \notin [a, b]$. These inclusion rules follow from order on the reals; induction on a rational expression proves the soundness of every resulting bound. All endpoints used here are rational, so no directed floating-point rounding is needed.

For the query calculation it suffices to use the larger box

$$\begin{gathered} c_\circ \in \frac{[318481, 318482]}{10^6}, \quad d_\circ \in \frac{[-1025233, -1025232]}{10^6}, \\ \rho \in \frac{[6019, 6021]}{10^5}. \end{gathered} \tag{46}$$

The first two inclusions follow from the root box; substitution in (20), with positive denominator, gives the third. Evaluate z_j from (23) and use

$$p_0 = (-1, 0)^T, \quad p_1 = (127/256, 1/16)^T, \quad p_2 = 16\rho(1 - c_\circ, -d_\circ)^T.$$

For a row with active index i, set $q = z_j + \alpha p_j$ and evaluate the following rational expressions by the inclusion rules:

$$(g_i - g_k)^T q \quad (k \neq i), \quad -\alpha g_j^T p_j, \quad \begin{cases} g_j^T z_j + \alpha g_j^T p_j/3 - g_i^T q, & \text{acceptance,} \\ g_i^T q - g_j^T z_j, & \text{rejection.} \end{cases}$$

For an acceptance also evaluate $\alpha g_i^T p_j$. Every expression in its row is strictly greater than the following positive rational lower bound, in the order of Table 2:

$$\frac{1}{10^4}(729, 2417, 145, 75, 5964, 2682, 1041, 221, 20). \tag{47}$$

The same interval operations give the value enclosures in that table. For the three base points, the designated active gaps are positive as well. Evaluating $3 \pm g_i^T q$ for every i, $2 - [q]_2$, and the designated active value $g_i^T q$ gives positive lower bounds for all base points and all nine queries. This proves (25). All inputs, expressions, interval rules and output bounds of this finite certificate are now specified. The strict active gaps also give differentiability on a neighborhood of each query. Together with the matrix identities, these calculations complete the proof of Lemma 4.

D Positive lower curvature through a perturbed cycle

Theorem 11 uses a different regularity balance from Theorem 10. Adding a positive quadratic changes the seed orbit, so exact homogeneity alone no longer supplies a trajectory. We first prove persistence on a stable manifold and then copy one entire perturbed orbit at all scales. The first step yields a global failure from a suitable matrix; the copying step makes bad starts approach the minimizer of one fixed objective; a final fixed change of coordinates makes their common initial matrix the identity. The price of retaining a positive lower curvature bound is the loss of second differentiability of three of the four branches at the minimizer. The line-search constants are any fixed pair $0 < c_1 \le 1/3$, $c_1 < c_2 < 1$ (an *admissible pair*), as in Theorem 10; the orbit, the objective and the starting points are chosen before the pair.

D.1 A strongly convex global example

A common positive quadratic leaves the active-branch geometry unchanged and makes every sublevel compact. It perturbs the gradients, however, so the homogeneous seed is no longer an exact orbit. Close to the corner, the quadratic gradient is small relative to the fixed slopes. The task is to turn this smallness into an exact infinite trajectory, with every rejection and acceptance retained.

Use the slopes g_0, g_1, g_2 and the function G of Section 6 and let

$$Q_\mu(x) = \tfrac{\mu}{2}\|x\|^2 + G(x), \qquad \mu > 0.$$

The vector

$$\bar g = \frac{(1, -16)^T}{257} = \frac{129}{257} g_0 + \frac{128}{257} g_1$$

is the least-norm point of $\operatorname{conv}\{g_0, g_1, g_2\}$: $g_0^T \bar g = g_1^T \bar g = \|\bar g\|^2 = 1/257$, whereas $g_2^T \bar g > 1/257$. Therefore the unique minimizer is

$$x_* = -\bar g/\mu, \qquad \min Q_\mu = -1/(514\mu). \tag{48}$$

At x_* the first two branches tie with distinct gradients, so the minimizer is nonsmooth. At zero the subdifferential is the nonstationary triangle of the original seed.

Proposition 1 (A perturbed bad orbit) *Let $\mu > 0$. For every neighborhood V of zero, some initial state (x_0, H_0), $H_0 \succ 0$, produces for every fixed pair $0 < c_1 \le 1/3$, $c_1 < c_2 < 1$ the same deterministic full-BFGS run on Q_μ, converging to zero with positive objective values decreasing to zero. Its accepted lengths repeat $1, 1/4, 1/16$, and its initial point and all accepted and rejected query points are differentiable and belong to V.*

The right object for persistence is the three-step return map after dividing out the shrinking scale. The homogeneous cycle becomes a fixed point on the boundary where that scale is zero. We will verify that it is hyperbolic and that its stable manifold enters the positive-scale region. This gives exact perturbed orbits; closeness of a finite floating-point trajectory would not. To prove the proposition, use normalized coordinates

$$x = \tau z, \qquad H = \tau M(\zeta, \eta), \qquad M(\zeta, \eta) = \begin{pmatrix} 1 & \zeta \\ \zeta & \eta \end{pmatrix}, \qquad \tau > 0.$$

The boundary seed is $P_* = (z_0, 0, 1, 0)$ in coordinates (z, ζ, η, τ). Keep the input τ fixed during three substeps. Starting with $(Z, M) = (z, M(\zeta, \eta))$, for $j = 0, 1, 2$ perform

$$\begin{aligned} \gamma_j &= g_j + \mu\tau Z, \qquad S = -\alpha_j M\gamma_j, \qquad Z^+ = Z + S, \\ Y &= g_{j+1} - g_j + \mu\tau S, \\ M^+ &= \left(I - \frac{SY^T}{Y^TS}\right) M \left(I - \frac{YS^T}{Y^TS}\right) + \frac{SS^T}{Y^TS}, \end{aligned} \tag{49}$$

with cyclic gradient indices and $\alpha = (1, 1/4, 1/16)$. For the resulting (Z_3, M_3), let $\hat\rho = (M_3)_{11}$ and define

$$\mathcal{T}_\mu(z, \zeta, \eta, \tau) = (Z_3/\hat\rho, (M_3)_{12}/\hat\rho, (M_3)_{22}/\hat\rho, \tau\hat\rho)\,. \tag{50}$$

The seed closure gives $\mathcal{T}_\mu(P_*) = P_*$ and $\hat\rho(P_*) = \rho$. All denominators are nonzero there. Hence the map is real analytic on an open neighborhood of P_*, using negative τ only to define this analytic extension.

Lemma 9 (Implementation and hyperbolicity) *On a sufficiently small neighborhood of P_*, positive-τ states follow exactly the three query lists of Table 2 under the actual algorithm for every admissible pair (c_1, c_2), and (50) is their return map. Its derivative at P_* has eigenvalues*

$$\rho^{-1}, \quad \rho^{-1}, \quad \lambda_s, \quad \lambda_u, \quad \rho, \qquad 0.038 < \lambda_s < 0.039, \quad 1.564 < \lambda_u < 1.565.$$

The stable eigenspace contains a vector whose τ coordinate is one.

Proof At a query τZ, the value divided by τ is $G(Z) + \mu\tau\|Z\|^2/2$ and the gradient is $g_i + \mu\tau Z$. The common quadratic does not affect which slope is active. All finite branch gaps and normalized search comparisons are continuous in the normalized state. The strict inequalities of Lemma 4, namely (24) at each acceptance and a value increase at each rejection, decide every test for all admissible pairs (c_1, c_2) at once; being finitely many, they persist on one neighborhood. The physical step and secant are exactly the ones in (49) after restoring the scale, proving agreement with the actual update and the entire search tree.

At $\tau = 0$ the matrix return is independent of z. Let J be the derivative at $(\zeta, \eta) = (0, 1)$ of its normalized shape map $(\zeta, \eta) \mapsto ((M_3)_{12}, (M_3)_{22})/\hat{\rho}$. The following rational enclosures specify the spectral certificate:

$$10^{10} J \in \begin{pmatrix} [11123986249, 11123986253] & [6459701025, 6459701029] \\ [7515015701, 7515015708] & [4905136882, 4905136890] \end{pmatrix}. \tag{51}$$

They are obtained by differentiating the rational substeps (49) at $\tau = 0$, then applying the interval rules of Appendix C to

$$c_\circ \in [318481427627118, 318481427627119]/10^{15},$$
$$d_\circ \in [-1025232119675749, -1025232119675748]/10^{15}.$$

Both intervals contain the defining root box. This is a finite rational derivative calculation: for a product use $D(ab) = a\,Db + b\,Da$, and for an inverse use $D(a^{-1}) = -a^{-2}Da$, with denominators bounded away from zero. Substitution of (51) in $p(\lambda) = \lambda^2 - \operatorname{tr}(J)\lambda + \det J$ gives

$$p(0.038) > 0, \qquad p(0.039) < 0, \qquad p(1.564) < 0, \qquad p(1.565) > 0.$$

These rational sign comparisons locate two distinct real roots and exhaust the degree-two polynomial; no numerical eigenvector is used.

At $\tau = 0$, $Z_3 = z + v(\zeta, \eta)$, with v the sum of the three normalized steps, and $\hat{\rho}$ is independent of z. The last coordinate of the return is $\tau\hat{\rho}$. Thus its full derivative has the block form

$$D\mathcal{T}_\mu(P_*) = \begin{pmatrix} \rho^{-1} I_2 & * & * \\ 0 & J & * \\ 0 & 0 & \rho \end{pmatrix}.$$

The blocks marked $*$ are finite derivatives of the displayed rational map and do not affect its spectrum. None of the five eigenvalues is zero or has modulus one. Moreover ρ is not an eigenvalue of the upper left four-dimensional block. Assigning last coordinate one and solving the remaining four linear equations gives a ρ-eigenvector. This proves the last assertion and the lemma. □

Proof (Proposition 1) The inverse function theorem makes the return map a local diffeomorphism. The classical local stable-manifold theorem provides a two-dimensional smooth stable manifold through P_*, tangent to the stable eigenspace; Dyatlov's local-coordinate formulation and graph-transform proof

apply after an adapted splitting and rescaling (*Notes on hyperbolic dynamics*, 2018 preprint, arXiv:1805.11660v1, Section 3.5, Theorem 3 and Remark 4; Section 3.1). By the last assertion of Lemma 9, this manifold contains points with arbitrarily small positive τ. Choose such a point whose forward return orbit stays in the implementation neighborhood and tends to P_*. Shrinking the neighborhood keeps $\hat{\rho} > 0$ and $M(\zeta, \eta) \succ 0$, so $\tau^+ = \tau\hat{\rho}$ preserves positivity. The actual intermediate matrices remain positive definite by positive curvature.

Let $w_m = (\hat{z}_m, \zeta_m, \eta_m, \tau_m)$ be the mth return of the chosen point w_0. The physical iterates are $x_{3m} = \tau_m \hat{z}_m \to 0$. Intermediate points and all queries are τ_m times bounded continuous functions of the normalized state, so they tend to zero too. Their normalized objective values stay positive by the seed's strict positivity. Armijo makes the values decrease to zero, which exceeds $\min Q_\mu$ in (48). More explicitly, each of the finitely many physical candidate maps, and the initial-point map $w \mapsto \tau z$, has the form $\Phi_q(w) = \tau q(w)$ and satisfies $\Phi_q(P_*) = 0$. Choose the normalized neighborhood inside $\bigcap_q \Phi_q^{-1}(V)$, then take a positive-scale local stable orbit that stays in this neighborhood. The initial point and every physical query lie in V. Neither the neighborhood nor the chosen point depends on (c_1, c_2), so the same run serves every admissible pair. This proves the proposition. The initial matrix is supplied by the stable-manifold point, not presumed to be identity or obtained by rounded forward shooting. □

D.2 An accumulation embedding with four strongly convex branches

The previous proposition places a bad orbit near its wrong corner, whereas the desired local counterexample needs bad starts near the minimizer. We therefore place shrinking copies around centers $\bar{x}_\ell$ converging to that minimizer. The spatial factor will be a_ℓ, and the objective factor a_ℓ^2, which preserves the same inverse matrix by (28). To accommodate the common quadratic, the three scalar branch slopes are shifted so that the quadratic cross term restores the original seed gradients inside every copy.

The scalar interpolation must have value a_ℓ^2 and derivative a_ℓ at its ℓth node, with an affine neighborhood wide enough to contain all seed queries after scaling. A monotone cubic ramp between these plateaux gives a convex function with Lipschitz derivative. The node slopes divided by the node positions stay of order one, so this construction need not, and will not, have a second derivative at zero.

Fix

$$\mu = 1/4096, \qquad c_* = (2, -32)^T, \qquad q_i = g_i - \mu c_*,$$
$$b_i = q_i^T c_*, \qquad r = 1/128, \qquad a_\ell = r^\ell.$$

Here $b_0 = b_1 = 1791/1024$ and $1.7 < b_i < 34$. For each of the two distinct values of b_i, set

$$t_b = \frac{1 - (1 + r)/b}{1 - r}, \qquad \varepsilon = 1/1024.$$

Use the same cubic ramp P_b and primitive J_b defined before (29), with this center and width. For $ba_{\ell+1} \le x \le ba_\ell$ put

$$\psi_b(x) = (ra_\ell)^2 + ba_\ell^2(1-r)\{r\xi + (1-r)J_b(\xi)\}, \qquad \xi = \frac{x/(ba_\ell) - r}{1-r}. \quad (52)$$

Set $\psi_b(0) = 0$, $\psi_b(x) = x^2/2$ for $x < 0$, and $\psi_b(x) = 1 + x - b$ for $x \ge b$.

On each positive interval,

$$\psi_b' = a_\ell\{r + (1-r)P_b(\xi)\}, \qquad \psi_b'' = P_b'(\xi)/b, \qquad 0 \le P_b' \le 768.$$

The identity $r + (1-r)(1-t_b) = (1+r)/b$ matches node values and derivatives. Their values are a_ℓ^2 and a_ℓ, respectively. The positive derivative is monotone and constant near every node. Its derivative slopes are uniformly bounded, so summing over intervals and taking a limit at zero proves that ψ_b is globally $C^{1,1}$, convex and coercive, nonnegative with unique zero at the origin. The constant-derivative lengths below and above ba_ℓ, divided by a_ℓ, are

$$b(1-r)(1-t_b-\varepsilon), \qquad b(1-r)(t_b-\varepsilon)/r,$$

both greater than $1/2$ for these values of b; the affine outer extension covers the upper side when $\ell = 0$. Consequently

$$\psi_b(ba_\ell + h) = a_\ell^2 + a_\ell h \qquad (|h| \le a_\ell/2). \quad (53)$$

At zero, the difference quotients of ψ_b' along ba_ℓ and $(b + 1/4)a_\ell$ equal $1/b$ and $1/(b + 1/4)$, respectively. Thus its second derivative at zero does not exist.

Define

$$\begin{aligned} f_i(x) &= \tfrac{\mu}{2}\|x\|^2 + \psi_{b_i}(q_i^T x) \qquad (i = 0, 1, 2), \\ f_3(x) &= \tfrac{\mu}{2}\|x\|^2 + x_2, \qquad F = \max_i f_i. \end{aligned} \quad (54)$$

Each branch is globally μ-strongly convex and $C^{1,1}$. For $i < 3$, choose a vector v_i with $q_i^T v_i = 1$. Restricting f_i to tv_i and subtracting its smooth quadratic term recovers $\psi_{b_i}(t)$, so these three branches are not twice differentiable at zero. An invertible linear change preserves this obstruction. For the first three branches, the gradient-Lipschitz constants are at most $L_i = \mu + \max\{1, 768/b_i\}\|q_i\|^2$. Direct substitution of the rational root bounds gives $L_0 < 439$, $L_1 < 446$ and $L_2 < 27$; hence 512 is a common constant, also covering the fourth branch. Since $F(x) \ge \mu\|x\|^2/2$ and $F(0) = 0$, the sublevels are compact and the minimizer is unique. The first three gradients at zero vanish, whereas the fourth is $(0, 1)^T$, so the minimizer is nonsmooth.

At a differentiable point of a finite maximum of L-smooth functions, all active gradients agree. Inactive branches stay inactive locally; taking the maximum of the active upper Taylor bounds therefore gives the local upper Taylor bound with constant L. Applying this with $L = 512$ and combining it with strong convexity proves

$$\mu I \preceq \nabla^2 F(x) \preceq 512 I \quad \text{whenever } F \text{ is twice differentiable.}$$

This does not require a globally Lipschitz gradient for the maximum itself.

D.3 Copying one entire orbit and fixing the initial matrix

Let $\bar{x}_\ell = a_\ell c_*$ and $C_* = 1 + \mu\|c_*\|^2/2$. For y in a fixed sufficiently small neighborhood of zero, (53) yields

$$F(\bar{x}_\ell + a_\ell y) = a_\ell^2\{C_* + Q_\mu(y)\} \qquad (\ell \geq 1). \tag{55}$$

Indeed, $|q_i^T y| \leq 1/2$ puts each old branch in its affine scalar region, and expansion of the common quadratic uses $q_i + \mu c_* = g_i$. The fourth branch is inactive there: its residual $a_\ell(-32 + y_2)$ is negative, while each old scalar residual is at least $a_\ell^2/2$.

Choose, by Proposition 1, one bad orbit $(x_k^{\text{seed}}, H_k^{\text{seed}})$ whose initial point and queries all lie in this neighborhood; it is the same for every admissible pair (c_1, c_2). For every $\ell \geq 1$, start at $\bar{x}_\ell + a_\ell x_0^{\text{seed}}$ with the same matrix H_0^{seed}. Induction on the actual query tree gives

$$x_k^{(\ell)} = \bar{x}_\ell + a_\ell x_k^{\text{seed}}, \qquad H_k^{(\ell)} = H_k^{\text{seed}}.$$

This is (27) with spatial factor a_ℓ and objective factor a_ℓ^2. The covariance calculation (28) preserves the matrix, while function differences and directional products acquire the same positive factor. Thus each rejection, acceptance and updated matrix is preserved. Equation (55) applies at the initial point and to every query because the whole chosen seed orbit lies in its domain. Thus $x_k^{(\ell)} \to \bar{x}_\ell$ and $F(x_k^{(\ell)}) \downarrow a_\ell^2 C_* > 0$. The three active gradients at $\bar{x}_\ell$ are $a_\ell g_i$, whose convex hull excludes zero. All query norms are at most $a_\ell(\|c_*\| + \sup_{\text{seed queries}} \|x^{\text{seed}}\|) \to 0$.

The seed supplied by the stable manifold has a positive-definite initial matrix H_0^{seed}, common to every copy. Identity initialization is obtained by changing coordinates once for the whole objective and the whole sequence of starts, rather than selecting a different objective for each start. Set $A = (H_0^{\text{seed}})^{1/2}$ and $\widetilde{F}(z) = F(Az)$. The coordinate identities

$$\widetilde{g} = A^T g, \qquad \widetilde{H} = A^{-1} H A^{-T}, \qquad \widetilde{s} = A^{-1} s$$

preserve both scalar line-search tests and (19); see also [1, Section 2]. They give $\widetilde{H}_0 = I$ for this one fixed transformation. Strong-convexity and smoothness constants become $\underline{\mu} = \mu\lambda_{\min}(H_0^{\text{seed}})$ and $\overline{L} = 512\lambda_{\max}(H_0^{\text{seed}})$. All locality and differentiability statements persist under the invertible linear map. This proves Theorem 11.

E Checking the hypotheses on actual runs

The examples below check the objective assumptions, comparison fields and full-BFGS histories together. They distinguish a function satisfying a convergence theorem from an infinite admissible run on that function, and from existence for every prescribed initial state. All norms in this appendix are Euclidean, and an admissible run has the meaning of Definition 1.

E.1 A nonsmooth certificate and a full-rank calibration

Proposition 2 (The mixed ridge, with an explicit history) *Let $F(u,v) = u^2 + |v|$ and $K_b = \{F \le b\}$, where $b > 0$. The function is finite and convex, its unique minimizer is zero, and K_b is compact. For every $0 < \eta \le b$, the following data satisfy all the conditions of Definition 2 on K_b:*

$$W_\eta(x) = xx^T, \qquad m_\eta = \eta^2, \quad C_\eta = 2b, \quad L_\eta = 0, \quad \delta_\eta = 1.$$

For $b = 2/3$ an actual infinite admissible run in K_b, with $(c_1, c_2) = (1/4, 1/2)$, is

$$x_k = \big(0, \tfrac{2}{3}(-\tfrac{1}{2})^k\big), \qquad g_k = (0, (-1)^k), \qquad H_k = \begin{pmatrix} 1 & 0 \\ 0 & 2^{-k} \end{pmatrix}, \qquad \alpha_k = 1.$$

It has $H_0 = I$ and converges to zero, although the smallest eigenvalue of H_k tends to zero.

Proof The objective assertions follow directly from $F(u,v) = u^2 + |v|$. At a differentiable band point, $g = (2u, \operatorname{sign} v)$ and $g^T x = 2u^2 + |v| \ge F(x) \ge \eta$, which gives visibility. Write $x = (u,v)$, $z = (r,w)$ and $y = \nabla F(z) - \nabla F(x)$. If v, w have the same sign, then

$$t = 2(r-u)^2, \qquad y^T z = 2(r-u)r, \qquad (y^T z)^2 \le 2F(z)t.$$

If their signs differ, put $S = |v| + |w| > 0$. Then

$$t = 2(r-u)^2 + 2S, \qquad y^T z = 2(r-u)r + 2|w|.$$

Cauchy–Schwarz gives

$$(y^T z)^2 \le t\left(2r^2 + \frac{2|w|^2}{S}\right) \le 2F(z)t \le 2bt.$$

Thus secant control holds for all differentiable endpoint pairs in K_b, in particular for the eligible ones. For an eligible pair, $s \ne 0$ and $z = x + s$ give $\Pi_s z = \Pi_s x$, so projected transport holds with equality. The displayed history has $s_k = -H_k g_k$, $y_k = -2g_k$, and

$$d_k = 2^{-k}, \qquad F(x_k) - F(x_{k+1}) = \tfrac{1}{3}d_k, \qquad t_k = 2d_k.$$

Consequently both line-search inequalities hold. The BFGS update leaves the first diagonal entry unchanged and replaces the second by 2^{-k-1}; all iterates are differentiable and nonstationary. This verifies the complete admissible history and the asserted limit. □

Proposition 3 (A full-rank certificate in every dimension) *For $F(x) = \frac{1}{2}\|x\|^2$ on $\mathbb{R}^n$, $n \ge 1$, every positive sublevel K_b is compact and zero is the unique minimizer. For $0 < \eta \le b$, Definition 2 holds with*

$$W_\eta(x) = I, \qquad m_\eta = 2\eta, \quad C_\eta = 1, \quad L_\eta = 0, \quad \delta_\eta = 1.$$

The field is positive definite of rank n. For any $x_0 \ne 0$, the history $x_k = 2^{-k}x_0$, $g_k = x_k$, $H_k = I$, $\alpha_k = 1/2$ is an infinite admissible run with $(c_1, c_2) = (1/4, 1/2)$, contained in $K_{F(x_0)}$ and converging to zero.

Proof Visibility is $g^T I g = \|x\|^2 = 2F(x) \geq 2\eta$. For any two endpoints, $y = s$, so $y^T I y = t$. For eligible pairs the constant field gives equality in projected transport. In the stated history the BFGS update fixes I because $y_k = s_k$. The actual decrease is $3d_k/4$, and the weak-Wolfe inequality holds with equality since $g_{k+1} = g_k/2$. Nonzero initial data give nonzero gradients at every finite index; the other assertions follow from the displayed geometric decay. □

E.2 Existence of differentiable weak-Wolfe steps

Lemma 10 (An existence criterion for admissible runs) *Let $F : \mathbb{R}^n \to \mathbb{R}$ be finite, convex and bounded below. Suppose that, for every differentiability point x and every $p \neq 0$, the set*

$$\{t \in \mathbb{R} : F \text{ is not differentiable at } x + tp\}$$

is countable. At any differentiable x with $g = \nabla F(x) \neq 0$, any $H \succ 0$ and any $0 < c_1 < c_2 < 1$, the direction $p = -Hg$ admits a positive step whose endpoint is differentiable and which satisfies strict Armijo and strict weak Wolfe. If, in addition, every differentiability point has nonzero gradient, an infinite admissible run exists for every differentiable x_0, every $H_0 \succ 0$ and every such pair (c_1, c_2).

Proof Put $\phi(t) = F(x+tp)$, $D = g^T p < 0$, and $\sigma(t) = (\phi(t) - \phi(0))/t$ for $t > 0$. Continuity and differentiability at zero give $\sigma(t) \to D$ as $t \downarrow 0$. Boundedness below gives $\sigma(t) > c_1 D$ for all sufficiently large t. Since $D < c_2 D < c_1 D$, continuity supplies a nonempty open interval on which $c_2 D < \sigma(t) < c_1 D$. Choose α there outside the countable exceptional set. The upper bound is strict Armijo. The supporting inequality at $x + \alpha p$ gives $\nabla F(x + \alpha p)^T p \geq \sigma(\alpha) > c_2 D$, hence strict weak Wolfe. Its positive secant curvature preserves positive definiteness under BFGS. Under the additional nonstationarity hypothesis these facts allow the step construction to be repeated indefinitely. □

The exceptional set concerns differentiability in the full ambient space, not just differentiability of $t \mapsto F(x + tp)$. The lemma proves existence of a sequence of acceptable steps; it does not assert that a prescribed line-search implementation finds them.

E.3 A simple family with curved switching

Proposition 4 (A two-quadratic family) *For $a, b, c > 0$, define*

$$F_{a,b,c}(u, v) = \max\{au^2 + bv^2 + v,\ cu^2 + bv^2 - v\}.$$

The function has unique minimizer zero and compact sublevels. It satisfies the hypotheses of Corollary 6 on every positive sublevel, and its two active branches at zero satisfy the affine-independence and positive-balance assumptions of

Corollary 7. For every differentiable initial point, every $H_0 \succ 0$ and every fixed $0 < c_1 < c_2 < 1$, an infinite admissible run exists. Every infinite admissible run converges in value and iterates to zero. The switching curve is the parabola $v = (c-a)u^2/2$, a line when $a = c$.

Proof Both branches have Hessian at least $2\min\{a, b, c\}I$, and their average is $(a+c)u^2/2 + bv^2$. Thus zero is the unique minimizer, sublevels are compact, and positive contours have no line segment. The two gradients differ on the tie curve: their second components differ by two. On that curve their determinant is

$$-2u\big(a + c + b(c-a)^2u^2\big).$$

It is nonzero unless $u = 0$, when the tie point is zero. At a smooth point affine criticality means stationarity, hence optimality by convexity. At a nonzero tie point the two linearly independent gradients have affine hull excluding zero. Therefore $\mathcal{C}_{\text{aff}} = \{0\}$; away from that set the switching is regular, and Corollary 6 applies on any positive sublevel. At zero the gradients $(0, 1)$ and $(0, -1)$ are affinely independent and balance with weights $1/2$.

Along any line through a differentiability point, the difference between the two branches is a polynomial of degree at most two, nonzero at the initial point. Hence there are at most two nondifferentiability points on that line. A differentiable stationary point would be the unique minimizer, but zero is not differentiable. Lemma 10 therefore gives the claimed run existence for every prescribed initial state and valid constant pair. □

Corollary 9 (Runs on the printed nonradial example) *For the objective of Theorem 9, every differentiable x_0 with $F(x_0) < 1/100$, every $H_0 \succ 0$ and every $0 < c_1 < c_2 < 1$ admit an infinite admissible run, and every such run converges to zero. In particular one may take $x_0 = (1/1000, 0)$, for which $F(x_0) = 1001/10^6 < 1/100$.*

Proof The objective is bounded below by $u^2 + v^2$ and has unique minimizer zero. At zero the distinct active gradients make it nondifferentiable, so every differentiable point has nonzero gradient. The restriction of each pairwise branch difference to a nonconstant line is a nonzero polynomial of degree at most two. Every line therefore meets the equality sets in finitely many points. Apply Lemma 10 and then Theorem 9. The value at the stated initial point follows by direct substitution. □

E.4 Why the run-existence qualification matters

Proposition 5 (A covered objective with no differentiable first step) *The function $G(u, v) = \max\{u, -u + v, -u - v\}$ is finite, convex, coercive and positively homogeneous, with unique minimizer zero. For every $\varepsilon > 0$, the point $x_0 = (\varepsilon, 0)$ is differentiable. With $H_0 = I$, its search ray has no differentiable positive-step endpoint satisfying weak Wolfe for any $c_2 < 1$. Thus no infinite admissible run has these initial data, although G belongs to the class of Theorem 6.*

Proof The function is a maximum of linear forms, and $G(u,v) \geq |u|$, $2G(u,v) \geq |v|$. These inequalities give coercivity and the unique minimizer. At x_0 the gradient is $(1,0)$, so the ray is $(\varepsilon - \alpha, 0)$. For $0 < \alpha < \varepsilon$ its gradient remains $(1,0)$, and weak Wolfe would require $-1 \geq -c_2$, which is false. For $\alpha \geq \varepsilon$ the endpoint is nondifferentiable: the last two branches tie with distinct gradients (at equality all three tie). These cases exhaust the positive ray. □

This is an obstruction to a differentiable infinite history from prescribed data, not an example of nonoptimal stagnation: the ray itself passes through the minimizer. The explicit histories above and the existence criterion specify which stronger non-vacuity conclusions accompany the convergence examples.

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